



Government of Odisha
Housing & Urban Development Department
3rd floor, Kharavel Bhavan, Unit-V, Bhubaneswar, PIN:751001

File No.: HUD-SANT-SCH-0026-2019 Letter No.: 22341/HUD, Date: 29.10.2024
From

Durgesh Nandini Sahoo, OAS (SAG)
Additional Secretary to Government &
Additional Mission Director, SBM (Urban)

To

The Commissioners of all Municipal Corporations
The Executive Officers of all Municipalities & NACs

Sub: Standard Operating Procedure (SOP) 3.0 for Municipal Solid Waste Management

Ref: This Department Letter No. 5993 dated 12.03.2024, No. 6003 dated 12.03.2024, No. 7086 dated 21.03.2024 and No. 9429 dated 30.04.2024

Madam / Sir,

In inviting a reference to the letters and subject cited above, I am directed to enclose a copy of the **"Standard Operating Procedure (SOP) 3.0 for Municipal Solid Waste Management"** for information and necessary action at your end.

The new "Enhanced Performance Indicator Framework" as indicated earlier in the letters referred above, accordingly forms a part of aforesaid SOP 3.0. The said "Enhanced Performance Indicator Framework" will be effective from 1st November, 2024 and the corresponding incentives as calculated on the basis of the said framework for the month of November, 2024 shall be payable during the month of December, 2024 and in the same manner thereof, until any further instructions in this regard.

Yours faithfully,

Enclosure: As above

Additional Secretary to Government &
Additional Mission Director, SBM (Urban)

Memo No. 22342 /HUD,

Date: 29.10.2024

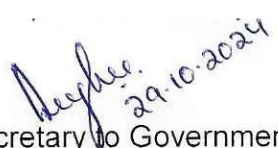
Copy along with copy of enclosure forwarded to PS to Special Secretary to Govt. & State Mission Director, SBM (U) for kind information of Special Secretary.

Additional Secretary to Government &
Additional Mission Director, SBM (Urban)

Memo No. 22343 /HUD,

Date: 29.10.2024

Copy along with copy of enclosure forwarded to the Collectors & DMs of all Districts/ Project Director, DUDA of all Districts / PMU, SBM (Urban) / Team Lead, Garima Cell / TSU-FSSM for information and necessary action.


Additional Secretary to Government &
Additional Mission Director, SBM (Urban)

**Standard Operating Procedure - 3.0
on
Municipal Solid Waste Management**

**Housing and Urban Development Department
Government of Odisha**

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Preface

The Housing & Urban Development (H & UD) Department, Government of Odisha assigns high priority to management of municipal solid waste in consonance with the provisions contained under the Solid Waste Management Rules, 2016. With the vision of “Swachha Odisha, Sustha Odisha” and based on the principles of circular economy, the State aims to recover “wealth” from “waste”. The State has adopted the decentralized model for solid waste management for all the ULBs except Berhampur Municipal Corporation having a centralised unit.

The decentralised model brings in less land requirement, transparency and cost efficiency compared to a conventional centralized model. Besides, it shifts the management of infrastructure from a contractor centric mode to a community driven model bringing ownership and livelihoods for the women members of the SHGs, Transgenders and Waste Pickers.

Consequent upon adoption of decentralised solid waste management system for collection, transportation, processing and safe disposal of municipal solid waste in the State, Micro Composting Centres (MCCs) and Material Recovery Facilities (MRFs) have been established which are usually co-located and are generating wealth out of waste and also found to have larger scope for which these institutions have been aptly termed as “Wealth Centres”.

The purpose of this Standard Operating Procedure (SOP) is to prescribe an updated process flow to manage solid waste by associating Community Partners, viz. women members of the SHGs, Transgenders and Waste Pickers in the Operation and Maintenance (O&M) of the “Wealth Centres” in the State of Odisha considering the learning after implementing the decentralised model adopting Standard Operating Procedure issued by this Department in Letter No. 13408 Dated 30.07.2019, Letter No. 19700 Dated 18.12.2020 and previously issued SOP 2.0 on Wealth Centre circulated vide the Housing & Urban Development Department Letter No. 21681 dated 20.12.2022. This revised **Standard Operating Procedure- 3.0 (SOP- 3.0)** is meant to supersede the previously issued SOPs.

Standard Operating Procedure - 3.0 on Municipal Solid Waste Management

1. Background:

The state has mandated the collection and transportation of segregated waste (wet and dry) from the households, commercial units, bulk waste generators including institutions and offices to the processing units. The wet waste (biodegradable) is processed at the Micro Composting Centre (MCC) into organic compost called "Mo Khata," and the dry waste (non-biodegradable) is sorted at the Material Recovery Facility (MRF) for onward sale of recyclables. The non-recyclables are disposed of scientifically. The revenue generated out of this endeavor flows back to the community in the form of incentives for the members of SHGs/Transgender Groups (TG) and Waste Picker Groups (WPGs) involved as Swachha Sathis, Swachha Supervisors (**henceforth to be referred & known as Plant Manager, MCC/ Plant Manager, MRF**) and Swachha Karmis in the municipal solid waste management.

In the entire decentralised sanitation value chain, the roles of Swachha Sathis and Plant Managers are critical towards ensuring the community link for sanitation services. The Swachha Sathis and Plant Managers belong to the community and are from the SHGs/TGs/WPGs, whose services primarily focus on changing behaviour of the citizens towards uptake and cooperation in sanitation services and enhancing the collection of User Fees.

This document deals with the Standard Operating Procedure (SOP) on collection, transportation, processing and safe disposal of municipal solid waste as well as the roles and responsibilities of key functionaries in the Solid Waste Management (SWM) value chain.

2. Objective:

The objective of this document is to establish a Standard Operating Procedure and clarity on the roles and responsibilities of key functionaries in the Solid Waste Management (SWM) value chain. The overall goal is to create awareness in the community about segregation of waste at the source, effective collection of waste from each household, commercial units, bulk waste generators including institutions and offices leaving no unit behind, ensure safe transportation of segregated waste directly to the processing units and to ensure processing of Solid wastes at Wealth Centres i.e. MCC and MRF and also safe disposal of non-recyclables and inert. Besides, this SOP also focuses on collection of the SWM user fees and the roles and responsibilities of stakeholders towards supporting other sanitation related surveys as and when required by the ULBs.

3. Generation of Municipal Solid Waste in Urban Odisha:

Sample survey on different aspects pertaining to generation of municipal solid waste taken up by the ULBs in respect of the households during August to December, 2019 reveals average waste to be 300 grams per person per day of a household in Urban Odisha on an average & wet waste and dry waste in the ratio of 60: 40. However, aforesaid ratio when taken up with waste generated by the commercial establishments, Bulk Waste Generators etc. may be revisited to be 50 : 50 which is also an indicative one.

Assumptions: Without delving deep into the numerical of waste generation and assuming the rate of waste generation and ratio of wet and dry waste as above the following assumptions may be made for designing operational strategy for the “Wealth Centres”.

Table 1: Generation of Municipal Solid Waste in Urban Odisha

Sl. No.	Assumption in generation of municipal solid waste in Urban Odisha	Quantity (Gram)
1	Average waste generation by a person per day	300
2	Average wet waste generation by a person per day (50% of total municipal solid waste)	150
3	Average dry waste generation by a person per day (50% of total municipal solid waste)	150
4	Average wet waste generation by a family (of 4.5 persons) per day (150 x 4.5)	675
5	Average dry waste generation by a family (of 4.5 persons) per day (150 x 4.5)	675
6	Total waste (Wet and Dry) generation by a family (of 4.5 persons) per day (300 x 4.5)	1350

3.1 Quantum of Waste Management:

Urban Local Bodies with variation in demography is to manage waste in proportion thereto as indicated below.

Table 2: Quantum of Waste to be Managed by ULB

Sl. No.	Type Of ULB	Population Range	Household Range [Population/4.5 (average number of member)]	Quantum of Waste to be managed per day (indicative) (@ 1350 gm. per household per day)	Number of 5 Ton MCCs and MRFs required
1	Municipal Corporation	> 3,00,000	>66,666	>89999100 g. (66,666 x 1350 g.) or >89999.100 Kg. or >89.999 Ton or 90 Ton	18+18

2	Municipality	25,000 to 3,00,000	5,555 to 66,666	7499250 g. (5,555 x 1350 g.) or 7499.250 Kg. or 7.499 Ton or 7 Ton to 89999100 g. (66,666 x 1350 g.) Or 89999.100 Kg. or 89.999 Ton or 90 Ton	2+2 To 18+18
3	NAC	10,000 to 25,000	2,222 to 5,555	2999700 gm. (2,222 x 1350 g.) or 2999.700 Kg. or 2.9 Ton to 7499250 g. (5,555 x 1350 g.) or 7499.250 Kg. or 7.499 Ton 7 Ton	1+1 to 2+2

4. Key Stakeholders across SWM value chain:

The solid waste management system in the state of Odisha has various stakeholders involved across its value chain. Each of the stakeholder has different role in different stages of SWM value chain as it is represented below.

Table 3: Responsibilities of stakeholders across the SWM value chain

Stakeholders	Source Segregation	Collection & Transportation	Processing	Disposal	Citizen awareness
Sanitation Expert					
Sanitation Intern					
Sanitary Inspector/ Health Inspector					
Swachha Sathis*					
Plant Managers					
Swachha Karmis* (Swachha Sabari Drivers and Helpers)					
Wealth Centre-Nodal Officer					
Swachha Karmis* at Wealth centre					

* are from SHGs/ Transgender Groups/ Waste Picker Groups

5. Swachhata Committee at the ULB level:

A “Swachhata Committee” shall be formed at the NAC/Municipality/ Zonal level in case of Municipal Corporations with the following composition:

Table 4: Members of ULB level Swachhata Committee

1	Commissioner or officer designated by Commissioner/Executive Officer	Chairperson
2	Sanitary Inspector (to be nominated by ULB)	Member
3	Representative of Mission Shakti Deptt. (to be nominated by MS Dept)	Member
4	Sanitation Expert (to be nominated by Commissioner/EO)	Member
5	City Mission Manager (to be nominated by Commissioner/ EO)	Member
6	City Engineer/ME/AE/JE (whoever is the senior most)	Member Convenor
7	Special Invitee (s) (if required as per the agenda)	Member

Special invitee is to be coopted depending on the agenda e.g., in case of a proposal for replacement of Swachha Sathi(s)/ Plant Manager(s)/SHGs/TGs/WPGs, the concerned Community Organizer and Ward Officer are to be coopted as the special invitees.

The Committee shall hold meeting once in every month to monitor and evaluate the performance in SWM.

6. Roles and responsibilities of the key stakeholders:

6.1 Roles and responsibilities of the ULB-level Swachhata Committee:

1. Approval of the selection of SHGs/TGs/WPGs for deployment of Swachha Sathis and Plant Managers.
2. Approval of rationalization of allotment of Households/Commercial units/Bulk Waste Generators including institutions and offices to Swachha Sathis, Plant Managers and Collection Vehicles.
3. Recommendation on the proposals for termination of Agreement between the ULBs and SHGs/TGs/WPGs / replacement of SHGs/TGs/WPGs, Swachha Sathis and Plant Managers to the Commissioner/ EO.
4. Periodic visits to the Wealth Centres and Wards for monitoring.
5. Attend the monthly Badakhana program.

6.2 Roles and Responsibilities of the Ward level Committee:

1. Monitor and certify the existence /non-existence of open dumping of Municipal Solid Waste called Garbage Vulnerable Points (GVP) in the Ward.
2. Support the Swachh Sathis in raising awareness on source segregation and collection of user fees.
3. Discuss status of Sanitation in the periodic Ward Level Meetings.
4. Monitor collection and transportation of waste from Commercial units/Bulk Waste Generators including institutions and offices.
5. Resolution of issues related to sanitation at the ward level.

6.3 Roles and Responsibilities of Nodal Officer, Wealth Centre:

Each ULB shall nominate a suitable Nodal Officer for Wealth Centre(s).

1. Monitor overall functioning of the wealth centre.
2. Evaluate the performance of the Swachha Sathi and Plant Manager on monthly basis.
3. Based on the evaluation, issue notices as detailed in point no 7.5 and 7.6.

6.4 Roles and Responsibilities of Sanitation Expert:

1. Ensure daily collection of segregated waste in the Wealth Centre assigned to them.
2. Check weekly attendance of Sanitation Intern/Wealth Centre In-Charge, Plant Managers, Swachha Karmis and Security Person of the Wealth Centre assigned to them.
3. Ensure elimination of GVP.
4. Monitor the monthly evaluation reports of Swachh Sathis and Plant Managers and facilitate the processing of incentives in time.
5. Ensure that payment of incentives to the Swachha Sathis/Plant Manager, MCC/ Plant Manager, MRFs/ Swachha Karmis are released by the 7th of every month.
6. Maintain repair and maintenance schedule of BoVs/LCVs and equipment at Wealth Centre.
7. Ensure daily reporting by Sanitation Intern/Wealth Centre In-Charge in the Ama Sahara app/MIS and monthly report on operation and management of MCC and MRF to ULB/State in the prescribed format.

8. Check and ensure the availability of consumables and other requirements at the Wealth Centre.
9. Undertake field visits to monitor on-ground performance of Swachha Sathis/Plant Managers.
10. Coordinate with Plant Manager of MCC/MRF for resolution of issues, if any.
11. Co-ordinate for servicing of machineries and equipment under Annual Maintenance Contract (AMC) by the agencies.
12. Ensure quality check, sample lab testing and packaging of compost (Mo Khata).
13. Ensure establishment of market linkages for the sale of “Mo Khata” and ensure timely supply of Mo Khata as per the indent received from departments/ bulk buyers.
14. Facilitate organising training and certification of wealth centre staff and SHGs/TGs/WPGs members.
15. Facilitate health check-up of wealth centre staff and SHGs/TGs/WPGs members.
16. Facilitate Insurance claim (if any) of the insured.
17. Support Swachha Sathis and Plant Managers in their activities as and when required.

6.4 Roles and Responsibilities of Sanitation Intern:

1. Verify and certify the performance of Plant Manager on waste collection from Commercial units and Bulk Waste Generators including institutions and offices.
2. Verify and certify the performance of Plant Manager on fleet management & monitoring the movement of Swachha Sabari and maintenance of the register.
3. Verify and certify the performance of Plant Manager on functioning of MCCs/MRFs, maintenance of records and cleanliness of the wealth centre.
4. Supervise the performance of the Swachha Karmis and ensure efficiency in processing of waste.
5. Ensure display of Signages and other IEC material at the Wealth Centre.
6. Ensure real time reporting in Ama Sahara App.

6.5 Roles and Responsibilities of Plant Manager, MCC / Plant Manager, MRF :

1. Ensure waste collection from Commercial Units & Bulk Waste Generators including institutions & offices.
2. Monitoring the movement of vehicles and maintain the register
3. Ensure regular maintenance and cleaning of drainage channel, periodic desludging of leachate tank/pit.
4. Ensure preparation of EM solution as per the SOP.
5. Weighing of incoming waste on weigh bridge.
6. Ensure unloading of incoming waste and maintenance of records
7. Ensure 100% segregation of wet and dry waste.
8. Ensure 100% sorting of dry waste received at MRF within a week's time.
9. Ensure safety and hygienic condition of MCC/MRF (Cleanliness of MCC/MRF, functional separate toilets for male & female)
10. HR management of Swachha Karmis at MCC/MRF ensuring each shift of Wealth Centre
11. Ensure wearing of Personal Protective Equipment (PPE) by Swachha Karmis while working
12. Timely packaging & lifting /sale of "Mo Khata" for MCC /recyclables & non-recyclables for MRF
13. Supervise the work of Swachha Karmis in the Wealth Centre (receiving and weighing the incoming waste and weighing of segregated waste, EM solution preparation and application, putting shredded waste mixing with EM solution in tubs, turning waste by following the prescribed procedures, testing, sieving and drying).
14. Facilitate minimization of downtime of machineries and preventive & responsive maintenance of machineries and equipment.
15. Facilitate repair and maintenance of BoVs/LCVs/Tricycles and electrical equipment/fixture, functioning of toilet, availability of water supply at wealth centre. Must raise indent to ULB and follow up with Sanitation Expert/Nodal Officer for timely action
16. Update information through Ama Sahara App, Books of account, Daily Information Board etc.
17. Ensure cleaning of the Wealth Centre and its premise before and after operations every day.

18. Inform the Sanitation Expert/Nodal Officer on operational issues including water & electricity supply and follow up till resolutions
19. Check daily attendance of Swachha Karmis and Security Persons.
20. Ensure proper coordination among the Swachha Sabari Drivers, Swachha Sathis, Zamadars, Sanitation Inspectors, Sanitation Experts and Sanitation Intern of the ULB;
21. Maintain a register to record any incidents/accidents at the Wealth Centre.

6.6 Roles and Responsibilities of Swachha Karmi at Wealth Centre (MCC & MRF):

6.6.1 Swachha Karmis at MCC:

1. Preparation of Effective Microorganism (EM) solution
2. Secondary segregation of waste
3. Weighment of segregated waste
4. Shredding wet waste through shredder
5. Preparation of compost tub to put in shredded waste.
6. Timely turning of waste in the tub
7. Drying & sieving of compost
8. Packaging of Mo Khata
9. After sieving, the un-processed waste to be further shredded and put in the compost tub for re-processing
10. Inert materials to be kept separately
11. Regular cleaning of waste processing area.

6.6.2 Swachha Karmis at MRF:

12. Secondary segregation and sorting of recyclable waste
13. Weighing of segregated recyclable waste
14. Storage of recyclable waste
15. Baling of non-recyclable wastes
16. Regular cleaning of waste sorting area

6.7 Roles and Responsibilities of Swachha Karmi (Swachha Sabari Driver and Helper):

1. Ensure cleanliness and readiness of the vehicle at the end of the day
2. Ensure that the Swachha Sabari starts collection at the designated time;
3. Ensure that the Swachha Sabari follows the route chart;
4. Ensure playing of IEC audio during the door-to-door collection;
5. Collection of segregated waste from door level and facilitate segregation of waste, if unsegregated
6. Make a note of Households, Commercial Units and Bulk Waste Generator including institutions and offices not providing waste or repeatedly providing unsegregated waste, intimate the Swachha Sathi and Plant Manager respectively.
7. Make a note of GVP and intimate Swachha Sathi for timely action.
8. Submit the daily collection sheet at the Wealth Centre
9. Report the breakdown of the vehicle to the Plant Manager;
10. Buying Mo Khata at discounted price from Wealth Centre and deliver to the Households against the indent received.
11. Report grievance, if any, from Households, Commercial Units and Bulk Waste Generator including institutions and offices during the time of collection of waste to the Swachha Sathis/Plant Managers.
12. Extend support in the operation and maintenance of Wealth Centre whenever required after completion of collection and transportation of waste within the working hours of the Wealth Centre.

6.8 Roles and Responsibilities of Swachha Sathi:

1. Ensure door-to-door collection of segregated waste -1) wet waste, 2) dry waste, 3) E-waste & Domestic Hazardous waste.
2. Visit all the tagged Households/Commercial Units/Bulk Waste Generators including institutions and offices and collect sanitation User Fee.
3. Ensure the elimination of GVP in the assigned area
4. Facilitate meetings on sanitation with Federation/SDAs, Ward Level Committees at least once in a month.
5. Conduct IEC campaign on sanitation as mandated by ULB
6. While collecting the waste, demonstrate segregation of waste to the households not practicing segregation at source

7. Ensure that the route map is affixed on the Swachh Sabari and is followed by the Swachha Sabari Driver. Maintain a visit register for Households, Commercial units and Bulk Waste Generators including institutions and offices.
8. Maintain a register for complaints and share with Sanitation Intern/Nodal Officer.
9. Support in sanitation related survey and activities.

7. Selection and deployment of Swacha Sathis, Plant Managers and Swachha Karmis from SHGs/ TGs/ WPGs:

1. The SHGs/TGs/WPGs shall be selected for the deployment of Swachha Sathis, Plant Managers and Swachha Karmis including Swachha Sabari Drivers and Helpers based on the eligibility criteria and selection process mentioned below.
2. The number of Swachh Karmis are to be decided by ULB based on the MCC/MRF's actual utilisation capacity of at least 1 Swachh Karmi per TPD utilisation capacity of MCC (up to 7 Swachha Karmis for 5 TPD) and 2-3 Swachh Karmis per TPD utilisation capacity of MRF.
3. ULB shall enter into an agreement with the selected SHGs/TGs/WPGs for providing Swachha Sathis, Plant Managers and Swachha Karmis for a period of 2 years with detailed terms and conditions, which shall be renewed for further period of 2 years based on their performance and willingness.
4. Termination/Replacement of the SHGs/TGs/WPGs and/or Swachha Sathis and Plant Managers shall be done by the Commissioner/ EO following due procedure as detailed in Point No 7.5 and 7.6 of SOP 3.0 on the recommendation of the Swachhata Committee.

7.1. Eligibility Criteria:

The eligibility criteria for SHGs/TGs/WPGs and the Swachha Sathis and the Plant Managers are as follows:

7.1.1. Eligibility Criteria for SHGs/TGs/WPGs:

- i. An impeccable past record both in terms of activity and financial management
- ii. Completed 1 year of existence
- iii. Must have an active Bank Account
- iv. Must have a Permanent Account Number (PAN)

- v. Must be strictly following Panchasutra
- vi. No criminal or anti-social track records.

7.1.2. Eligibility Criteria for Swachha Sathi and Plant Manager:

- i. Should have been a member of the SHGs/TGs/WPGs at least for last one (1) year
- ii. Should have passed Class X or above for Swachha Sathi and Class XII or above for Plant Manager
- iii. Should have a Bank Account in their name
- iv. Should have a Smart Phone in their name
- v. No criminal or anti-social track records.

7.2. Selection Process:

The ULBs shall select the SHGs/TGs/WPGs through the following process:

1. The ULB shall float an EoI for inviting applications from SHGs/TGs/WPGs for deployment of Swachha Sathis, Plant Managers and Swachha Karmis
2. The EoI shall be published by ULB in the ULB's official website and shall be displayed in the Notice Board of ULB, ICDS Office and Ward Office(s)
3. The EOI shall be opened in a ULB level Swachhata Committee meeting called for the said purpose.
4. The Minutes of meeting shall be prepared and filed for necessary action and future reference.
5. The committee will scrutinise and assess all the applications received in response to the EOI based on the eligibility criteria mentioned above and finally select the SHGs/TGs/WPGs. The ULB level Swachhata Committee reserves the right to reject the application submitted by SHG/TG/WPG if not satisfied with credibility and past record of the group.
6. An Agreement shall be signed between the elected SHG/TG/WPG and the ULB for deployment of Swachha Sathi/ Plant Manager/ Swachha Karmi
7. The SHG/TG/WPG shall nominate the eligible member for Swachha Sathi/ Plant Manager / Swachha Karmi as the case may be.
8. The contact details of the selected Swachha Sathi/ Plant Manager and the respective SHG/TG/WPG shall be displayed in the respective Ward Office (s)

7.3. Execution of Standard Agreement:

1. The ULB shall enter into an Agreement with the SHG/TG/WPG for deployment of the Swachha Sathis/Plant Managers/ Swachha Karmis.

2. The Agreement shall be in force for a period of two years from the date of entering into the Agreement unless otherwise decided by ULB. The Agreement, may be extended for a further period of one year subject to satisfactory performance by the SHG/TG/WPG/Swachha Sathis/ Plant Managers.
3. The Agreement shall contain the details of the roles and responsibilities of the Swachha Sathis/Plant Managers/ Swachha Karmis.
4. The Agreement shall also contain the details of the specific area assigned to each of the Swachha Sathis / Plant Managers, who shall perform their functions within the assigned area only. However, they can also perform the functions in an adjacent area temporarily in case of absence of a Swachha Sathis/Plant Managers subject to approval by the ULB. Accordingly, they will be eligible for the incentives for the said period.
5. The Agreement shall mention about the applicable incentives to be paid to the Swachha Karmis.
6. The Agreement for Swachha Sathis/Plant Managers shall mention the prescribed incentives based on performance indicators mentioned in this document.
7. The incentives payable shall be deposited by the ULB directly to the Bank Account of the Swachha Karmi / Swachha Sathi / Plant Manager.

7.4. Capacity Building of the selected SHGs/TGs/ WPGs:

To build, strengthen or upgrade the skills and capacities of the deployed Community Partners under SWM, the department has designed and rolled out the training modules, self-learning training guide books and information, ready reckoners for different stakeholders. The ULBs shall be responsible for ensuring that the training programmes are organised with support from the resource organisations and agencies authorised by the H & UD Department.

7.5. Termination of Agreement:

The Agreement may be terminated by either party under the following circumstances:

1. In the event the SHGs/TGs/WPGs are found to be violating the terms and conditions of the agreement or found to be not adhering to the terms and conditions, the same shall entail for termination of the Agreement with the Group by giving a notice in writing of not less than 30 days. During this period, the SHGs/TGs/WPGs shall be provided due opportunity to take corrective measures within 15 days from the date of notice with further extension on request in writing, not exceeding 7 days' time if required.

2. In case SHGs/TGs/WPGs expresses unwillingness/unable to continue with the Agreement, then the SHGs/TGs/WPGs shall have the right to get withdrawn from the Agreement on giving a 30 days' notice in writing to ULB.
3. In case of dissolution of SHGs/TGs/WPGs, Agreement shall stand terminated on receipt of either a copy of the resolution or intimation stating the dissolution of SHGs/TGs/WPGs.
4. The termination of the SHGs/TGs/WPGs shall be done by the Commissioner/ EO on the recommendation of the ULB level Swachhata Committee.

7.6. Replacement of Swachha Sathi/ Plant Manager:

1. Swachha Sathi may be replaced by the ULB by following due procedure as mentioned in the illustration given below on the ground of non-performance by them in two parameters i.e., User Fee Collection and GVP Free continuously for a period of 3 months.
2. Plant Manager may be replaced by the ULB by following due procedure as mentioned in the illustration given below on the ground of non-performance by them in all the parameters as detailed in the Incentive Structure mentioned in Table 3 continuously over a period of 3 months.
3. The names of the non-performing Swachha Sathis/Plant Managers will be notified to the concerned SHGs/TGs/WPGs in writing by the Nodal Officer of the Wealth Centre after observing their performance for a month. This shall be treated as the first notice. Based on the notice, handholding support by the Nodal Officer shall be provided to the Swachha Sathis/Plant Managers for improvement in their performance in the following month. If the non-performance continues even in the second month, the concerned nodal officer shall issue a written show cause notice to the SHGs/TGs/WPGs in the beginning of the third month and direct them to take needful action for improvement in the performance of the Swachha Sathi/ Plant Manager. This shall be treated as the second notice. Consistent non-compliance for the third month will lead to replacement of the Swachha Sathi/ Plant Manager. The Nodal Officer of the Wealth Centre shall report to the Commissioner/ EO regarding the continuous non- performance of the Swachha Sathi/ Plant Manager and request for replacement. The meeting of the Swachhata Committee shall immediately be convened and the report of the Nodal Officer shall be placed before the committee. Based on the recommendation of the Committee, the Commissioner/ EO shall immediately issue 7 days' written notice to the SHGs/TGs/WPGs for replacement of the Swachha Sathi/ Plant Manager.

4. All notices shall have four (4) copies, of which one copy shall be shared with SHGs/TGs/WPGs, one copy each to be displayed at the MCC/MRF and ULB notice board, and one copy is to be kept in the SHGs/TGs/WPGs file maintained at the ULB.

An Illustration of the replacement process

1	Non-performance of Swachha Sathi/ Plant Manager observed during the month of January	Notice by Nodal officer to the SHGs/TGs/WPGs by 7 th February
2	Continued non-performance of Swachha Sathi/Plant Manager observed in the month of February	Show cause by Nodal officer to the SHGs/TGs/WPGs by 7 th March
3	No improvement in the performance of Swachha Sathi/ Plant Manager observed in the month of March	Shall be reported by the Nodal Officer to the Commissioner/EO by 7 th April to place it before the Swachhata Committee within three working days.
4	Replacement of the Swachha Saathi/ Plant Manager	Based on the recommendation of the Swachhata Committee, Commissioner/ EO shall, issue, within two days of receipt of such recommendation, seven days notice to the SHGs/TGs/WPGs for replacement of the concerned Swachha Saathi/ Plant Manager.

5. SHGs/TGs/WPGs shall propose the name of a suitable replacement of the non-performing Swachha Sathi/ Plant Manager meeting the eligibility criteria within a period of seven working days with a written intimation to the ULB communicating the details of the Swachha Sathi/ Plant Manager. Upon acceptance or rejection by the ULB, the same shall be communicated to the Group. In case of acceptance, the selected candidate has to be deployed within two days of receipt of such communication. In case of rejection by the ULB, the

group shall provide another suitable member within three working days. Failure to provide suitable member within the stipulated time will make the Agreement with the SHGs/TGs/WPGs liable for termination and ULB may take necessary steps for deployment of Swachha Sathi/ Plant Manager, as the case may be, from any other suitable SHGs/TGs/WPGs selected following due procedure.

6. The ULB will impart capacity building training to the newly deployed Swachha Sathis/ Plant Managers for a minimum of two days before commencement of the work.
7. In case of any change in the number of Swachha Sathi/ Plant Manager, the Agreement shall be amended to reflect the updated number and details of Swachha Sathi / Plant Manager deployed in the MIS data sheet, Ama Sahar App, SHG/TG/WPG file maintained at the ULB level, HR record maintained.
8. Failure on the part of the Nodal Officer of the Wealth Centre in notifying the SHGs/TGs/WPGs on the non- performance of Swachha Sathi/ Plant Manager shall be construed as serious lack of performance on the part of the Nodal Officer and disciplinary action shall be taken by the Municipal Commissioner/ Executive Officer against the Nodal Officer for the same.

8.Collection and Transportation of Segregated Waste:

8.1. Collection of Segregated Waste i.e. Wet and Dry Waste:

Collection of “segregated waste” from the Households, Commercial Units and Bulk Waste Generators including institutions and offices being the fulcrum of success of this model, due diligence needs to be applied and all out efforts should be made to collect segregated waste, i.e., wet and dry waste collected separately from the Households, Commercial Units and Bulk Waste Generators including institutions and offices on daily basis. Swachha Sathis must continuously educate, motivate and take up appropriate Information, Education Communication (IEC) and Behavioural Change Communication (BCC) activities to ensure the same.

It may be ensured that horticulture, parks and garden wastes are collected separately in the service area of the MCC and processed in the Parks and Gardens. Necessary appropriate arrangements may be made to compost them.

8.2. Collection of domestic hazardous waste and E-waste:

Domestic hazardous waste and e-waste are collected on a designated day of every week and deposited at MRF in Wealth Centre for channelizing to the agencies identified by the ULB.

8.3. Use of BoV/LCV- “Swachha Sabari”:

In order to facilitate the collection with ease and dignity, Battery operated Vehicles (BoVs) (with separate chambers for the wet and dry waste) has been introduced through Government e-market (GeM Portal) has been introduced through Government e-market (GeM Portal) gradually phasing out Push Carts, Tri-cycles etc. Besides, LCVs are also being used for collection of segregated wastes from the households as well as bulk waste generators including institutions and offices.

The BoV utilised for door-to-door collection are designated as “**Swachha Sabari**” as these are smokeless and environment friendly. All the BoVs/LCVs used in collection of segregated waste from the Households, Commercial Units and Bulk Waste Generators including institutions and offices should reach a Households, Commercial Units and Bulk Waste Generators including institutions and offices within a specified time, with a soothing miking arrangement (Volume must be within the permissible limit) to inculcate confidence among the waste generator that the waste shall be collected every day on time.

8.4. Rationalisation of allotment of households/commercial units/bulk waste generators including institutions and offices to Swachha Sathis:

1. Each Swachha Sathi is to be tagged to around 600 Households/Commercial units/ Bulk Waste Generators including institutions and offices;
2. Deviation in the number of HH tagged should not normally exceed +- 25% (i.e. Tagging 450-750 HHs/Commercial units/ Bulk Waste Generators including institutions and offices to one Swachha Sathi). For example, if a ward has 700 Households/Commercial Units/ Bulk Waste Generators including institutions and offices, then a Swachha Sathi can be tagged to 700 Households/Commercial Units/ Bulk Waste Generators including institutions and offices together for easy coordination.
3. The number of Swachh Sathis to be tagged shall be determined by dividing the number of Households/Commercial Units/ Bulk Waste Generators including institutions and offices by 600. Any decimal value is to be rounded off (For example: If there are 1400 HHs, then the total Swachha Sathi required will be $1400/600 = 2.3$ which is rounded off to 2).
4. ULB shall tag one BoV to one Swachha Sathi as far as possible.

8.5. Tagging of BoVs/LCVs to Households, Commercial Units and Bulk Waste Generators including institutions and offices:

1. Based on the Digital Door Numbering (DDN) Data, BoVs/LCVs used for collection and transportation of waste shall be tagged to Households/Commercial Units/Bulk Waste Generators including institutions and offices.
2. One BoV shall be tagged to around 600 Households/Commercial Units/Bulk Waste Generators including institutions and offices and one LCV shall be tagged to 1000 Households/Commercial Units/Bulk Waste Generators including institutions and offices.
3. Any deviation in the number of Households/Commercial Units/Bulk Waste Generators including institutions and offices tagged should not normally exceed $\pm 25\%$ (i.e. between 450-750 HHs for one BoV and 750-1250 HHs for LCVs). For example, if a ward has 700 Households/Commercial Units/ Bulk Waste Generators including institutions and offices then BoV can be tagged to 700 Households/Commercial Units/Bulk Waste Generators including institutions and offices together for easy coordination.
4. Each BoV/LCV will have a serial number, which along with Ward No. is to be written on Swachha Sabari i.e. BoVs and LCVs.

8.6. Alternative arrangements during non-functionality of the BoVs/LCVs:

1. In case of non-functionality of the BoV/LCV, the ULB will make immediate alternative arrangements for collection and transportation of waste. For example, provide alternative or reserve vehicle and in absence of alternative vehicle, collect the waste in two bins (one for wet waste and the other for dry waste) mounted on a tricycle from the Households/Commercial Units/Bulk Waste Generators including institutions and offices and transfer the collected waste to the BoVs/LCVs serving the nearby area.
2. Concerned Swachha Sathi will supervise the collection and transportation of waste to the BoVs/LCVs serving the nearby area.

8.7. Tagging of Plant Managers to individual MCC/MRF:

One Plant Manager shall be tagged to the MCC or MRF. Plant Managers will focus on:

- a) Segregated waste collection from bulk waste generators including institutions and offices by the Swachha Sabaris

- b) Fleet management and monitoring of collection vehicles - In case of co-located MCC and MRF, the number of vehicles attached to the Wealth Centre shall be equally distributed among the Plant Manager attached to the MCC and MRF.
- c) Regular maintenance, cleaning and hygiene of MCC/MRF.
- d) Human Resource Management at MCC/MRF.
- e) Timely sale/lifting of Mo Khata /that of recyclable and non-recyclables in MCC and MRF respectively.

8.8. Tagging of Plant Managers to Commercial Units/ BulkWaste Generators including institutions and offices;

- 1. Commercial Units and Bulk Waste Generators including institutions and offices to be identified and tagged to Plant Managers as per their area of operations.
- 2. A monitoring checklist is to be prepared for all the Plant Managers and provided to Plant Managers and Swachh Sabari Drivers.
- 3. The Swachha Sabari Drivers shall update the checklist and submit to Plant Managers daily.
- 4. The Plant Manager shall review the checklist and take necessary action for Commercial units, Bulk Waste Generators including institutions and offices.
- 5. The Plant Manager shall maintain the record for all the Commercial Units, Bulk Waste Generators including institutions and offices in a register.
- 6. The register is to be reviewed by Sanitation Intern every month.

8.9 Route Rationalization and Transportation of Waste:

Route rationalization is an exercise which helps in optimizing the waste collection process and to ensure 100% collection coverage. This exercise includes identification of waste generators, identification of routes (lanes) for door-to-door collection of waste.

Route Mapping/ Rationalization help in the following -

- 1. Identifying areas to be covered under Door to Door (D2D) waste collection
- 2. Investigate the extent of source segregation of waste
- 3. Identifying existing gaps in planning and engagement of vehicles in D2D waste collection
- 4. Identifying various underlying issues in solid waste collection such as waste burning spots, littering spots, etc.
- 5. Ensuring regularity of D2D services
- 6. Strengthening monitoring of D2D waste collection services
- 7. Efficiently managing the D2D waste collection services.

8.10 Route Mapping:

Route mapping can be done manually using local area maps/ sketches. The routes of vehicle are mainly dependent on the following factors:

1. Route for waste collection
2. Type, capacity and number of the vehicle
3. Number of points of intersect/gates/waste collection points

The process of route rationalization includes mapping of existing vehicle routes, households, road/streets, and other landmarks, planning of trips for each route and identification of points of waste collection or gates.

8.11 Pre-requisites for Route Rationalization:

The following are the prerequisites for the route rationalization process:

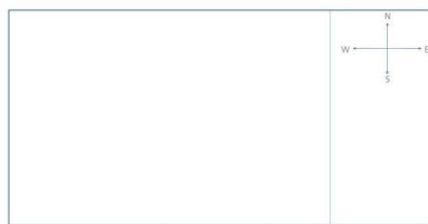
- a. Waste quantification must be carried out to manage and optimize the route rationalization process
- b. Knowledge of the existing waste collection routes and ward boundaries
- c. Waste carrying capacity of the vehicles
- d. Location and distance of the wealth centres from respective areas.

Information Box

Steps for route rationalization

STEP-1

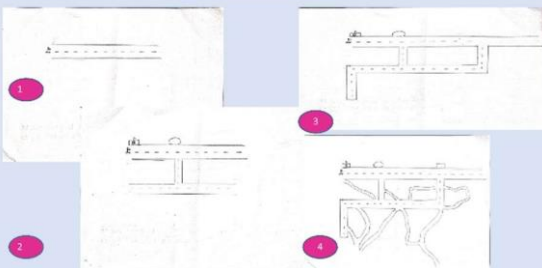
First put the directional symbol on the upper right corner of drawing sheet



STEP-2 Identify legends and list them

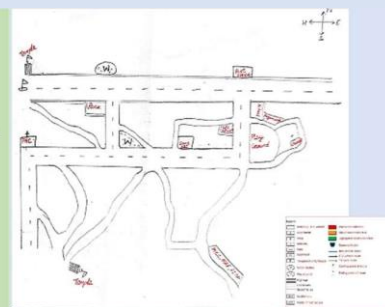
Legend	
Individual Household	Waste not collected
Apartment	Mixed waste collected
Shop	Segregated waste collected
Industry	Community bin
Park	Boat vehicle route
Aganwadi	LCV vehicle route
Temple/Church/Mosjid	Tri cycle route
Water bodies	Starting point of route
Play ground	Ending point of route
Highway	
Link Roads	
Street/ lanes	
Institutions (Write the name of institutions in the box - bank, post office, town hall, community hall, school, university, colleges, etc.)	
Public Infrastructure (Write the name of infrastructure in the box - SAC, MB, DTPS, PHC, etc.)	

STEP-3 Start marking roads, lanes, streets from the starting point of the route



STEP-4

Start marking landmarks as defined in the legend such as Banks, Post office, Waterbodies, Playground, Religious places, etc.



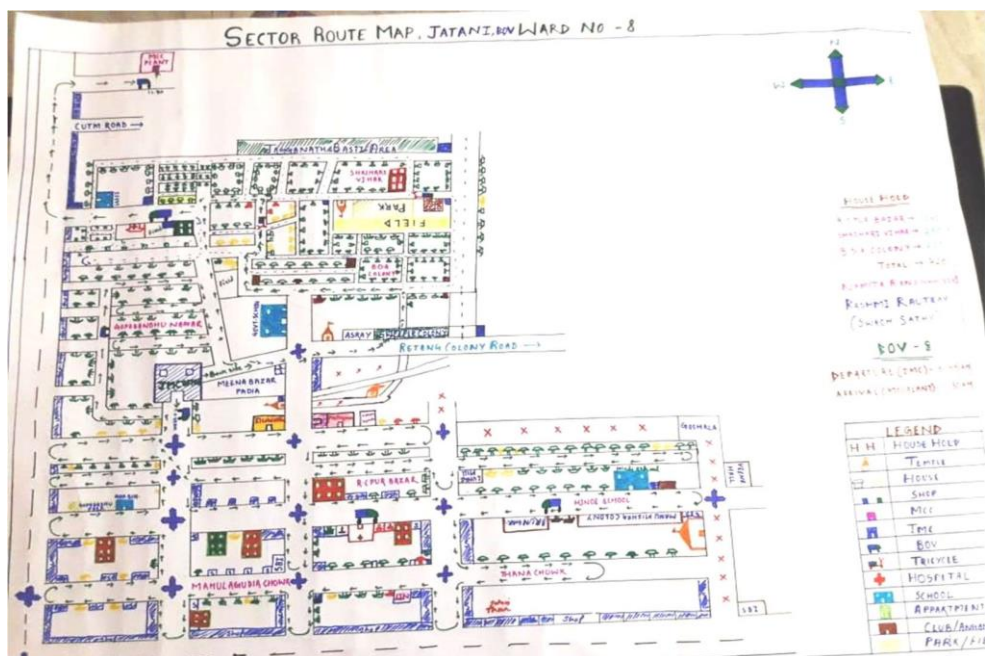
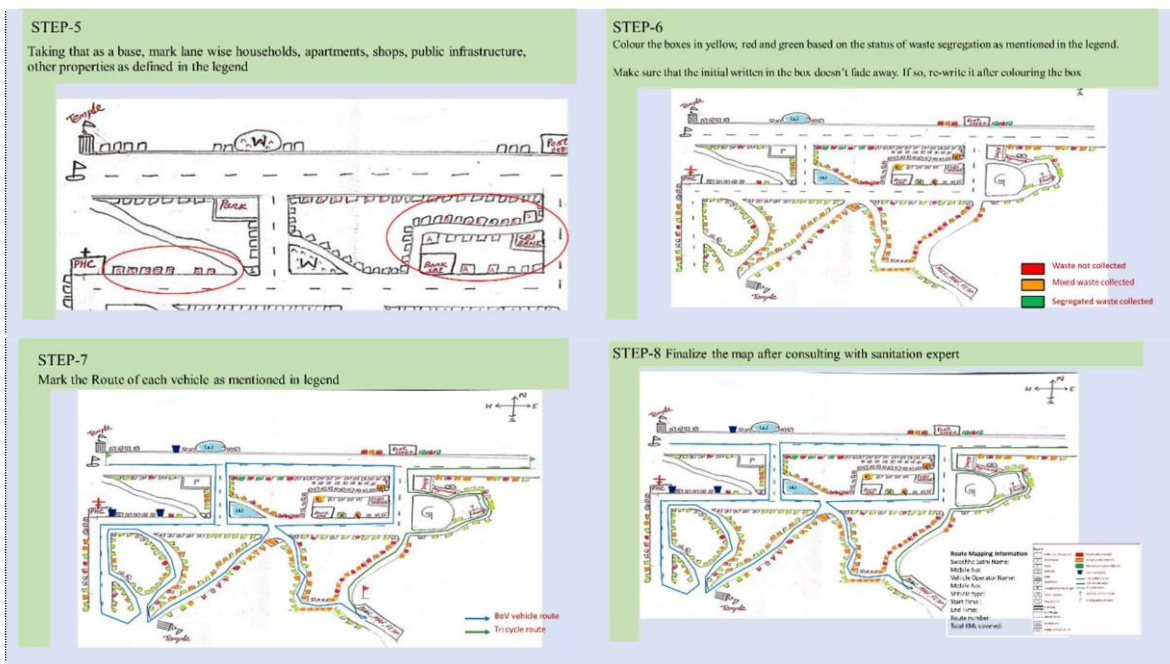


Figure 1: Route Map of an Area for Collection of Waste

Route Maps are prepared by Swachha Sathis and fixed in the vehicles along with target of solid waste to be collected by each vehicle. A register with title “Vehicle Deployment and Waste Collection Register” shall be maintained at every wealth centre.

9. Processing of Waste:

9.1. Wealth Centres:

The waste generated by households, commercial units and Bulk Waste Generators including institutions and offices are safely transported to the Wealth Centres (MCC and MRF) for processing and onward safe disposal of non-recyclables.

9.2. Weighing and receiving waste at Wealth Centre:

Weighing and receiving of waste comprises of the following steps:

- a. The source segregated waste from the households and bulk consumers is transported to the designated wealth centres.
- b. The waste-loaded vehicle (BoV/LCV) is weighed at the weigh bridge at the wealth centre
- c. The wet waste chamber is unloaded at the at MCC and the vehicle is weighed to ascertain the weight of wet waste*
- d. The dry waste is unloaded at the receiving yard of the MRF.
- e. Then the empty vehicle is weighed.
- f. The weight of the dry waste* is ascertained by simple calculation.
- g. The process is repeated for each trip and captured in a register.

The cumulative data for wet and dry waste for all trips for the day is calculated by adding the weight of wet and dry waste for each trip in the day and entered in Ama Sahara App.

*Weight of loaded vehicle – Weight of vehicle after unloading wet waste = Weight of Wet Waste received
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**Weight of vehicle after unloading wet waste (vehicle with dry waste) – Weight of empty vehicle = weight of dry waste received

9.3. Capacity of MCCs and MRFs:

MCCs and MRFs have been designed for processing of municipal solid waste ranging from one and half Ton per day (TPD) to five Tons per day (TPD) to be adopted suitably by the ULBs taking into consideration various factors such as availability of land, quantum of waste generated etc.

9.4. “Cluster” approach in managing waste:

The Table below indicates at a glance the quantum of waste that can be processed, requirement to be fulfilled by one Wealth Centre i.e. one cluster having one unit of MCC and one unit of MRF, as these two are co-located and therefore to be considered as a “Cluster”.

Table 7: Coverage by One Wealth Centre (Cluster with one MCC & one MRF)

Sl. No.	Capacity of one MCC (TPD)	Population to be covered (Capacity of one MCC ÷ Processing of 150 g. wet waste per person)	Households to be covered (With 4.5 persons on an average in each HH)
1.	5	33,333 (5000000 g. ÷ 150 g.)	7,407 (33,333÷4.5)
2.	4	26,666 (4000000 g. ÷ 150 g.)	5,925 (26,666÷4.5)
3.	3	20,000 (3000000 g. ÷ 150 g.)	4,444 (20,000÷4.5)
4.	2	13,333 (2000000 g. ÷ 150 g.)	2,962 (13,333÷4.5)
5.	1.5	10,000 (1500000 g. ÷ 150 g.)	2,222 (10,000÷4.5)

10. Micro Composting Centre (MCC):

10.1. Conceptual Design:

Micro Composting Centres (MCC) is designed for “composting” biodegradable waste (wet waste) which means a controlled process involving microbial decomposition to generate organic compost through aerobic composting.

10.2. Area required for MCC:

As per the provisions contained under rule 12 (a) of the SWM Rules, 2016 the Collector of the District shall facilitate identification and allocation of suitable land (including advance possession) at various locations in the ULB.

Table 8: Criteria for space requirement for MCC

Processing capacity of biodegradable waste in TPD	No. of Tubs	Size of each Tub	Required area for processing centre in Sq.m.*
5	14	3.9m x 1.8m x 1.0m	600
4	14	3.0m x 2.0m x 1.0m	410
3	14	3.0m x 1.5m x 1.0m	360
2	14	2.4m x 1.2m x 1.0m	260
1.5	14	2.0m x 1.2m x 1.0m	200

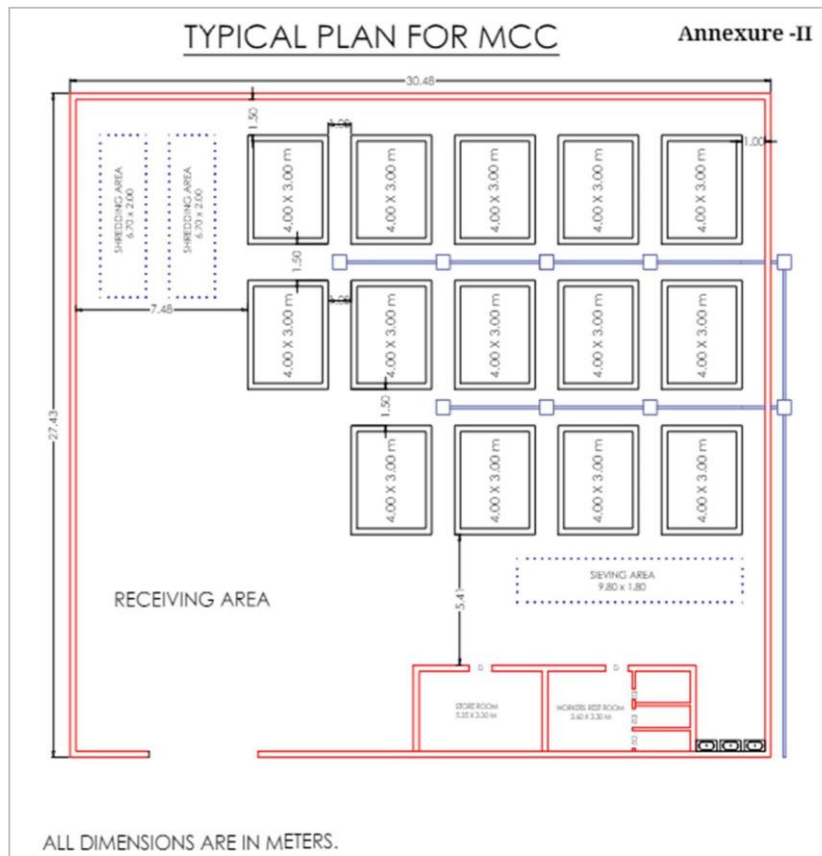
*Facilities for provisioning of shredding machine with conveyor belt, sieving machine, weighing machine, office room, washrooms (for men and women separately), Lounge, Locker facilities, assured source of water supply along with adequate drinking water supply system etc. should be provided.

Source: (Standard Operating Procedure for Decentralised Solid Waste Management in Odisha, 2019 issued by this Department)

Suitable location and capacity of Micro Composting Centre (MCC) may be determined based on the extent of land available. For example, a land parcel of 410 square meter can be used for establishing MCC of 4 Tons per Day (TPD) capacity. Based on the capacity of MCC, the coverage area comprising ward(s) can be determined. For

example, a 1.5 TPD MCC can get waste from 10,000 people that is about 2,220 households. Therefore, the coverage area of 2,220 households shall be delineated. (Refer calculation given below)

Additional MCCs, if required, may be taken up immediately for construction and operationalisation. All equipment required should be procured and installed. It should be ensured that, the ULB has the infrastructure in position to process all the municipal solid waste generated in it.



Calculation:

Capacity of MCC = 1.5 TPD = 15,00,000 grams per day

Per capita waste generation (assumed) = 300 grams per day

Bio-degradable per capita waste generation (assumed 50%) = 150 gram per day

The MCC can cover (15,00,000 grams per day / 150 gram per day) = 10,000 population

10,000 population = 10,000/4.5 = 2,222 Households (Assumed population per Household = 4.5)

Hence, a 1.5 TPD MCC can cater to 10,000 population that is about 2,220 households.

10.3. Layout Design of MCC:

Figure 4: Model Layout design of MCC

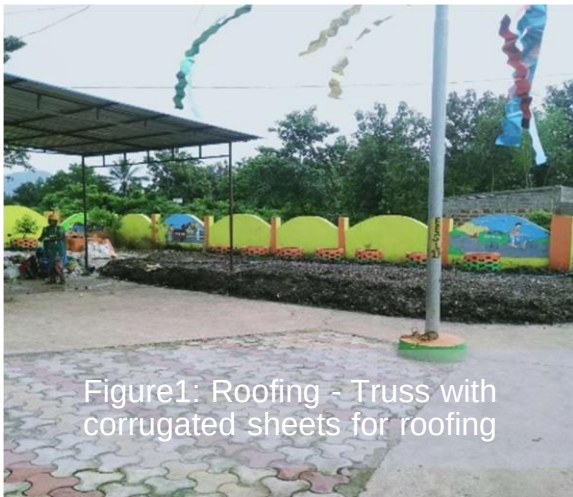


Figure1: Roofing - Truss with corrugated sheets for roofing



Figure 2: Truss with sheets installed



Figure 3: Side walls with mesh and wall paintings



Figure 4: Flooring - Concrete floor (anti-skid) with proper slope at the main entrance of the MCC

- Floor of each tub shall be sloped to one side and have a drain point with jalli to collect leachate.
- Leachate drain points from each tub shall be connected to underground pipes to form a leachate drainage network and connected to a chamber for safe collection.
- The accumulated leachate will be rich in bacterial consortium and can be used as inoculum. The leachate can be sprinkled a little on the waste in tubs. But too much of liquid may block the pores that will create anaerobic condition in the tubs.
- Each tub will have sufficient number of holes on the side walls. The holes will be connected through pipes and will have a cowl installed on the top to help proper ventilation.
- There shall be adequate space for movement (approximately 1meter) around each tub.

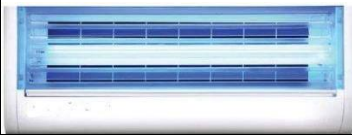
Table 9: Tub dimensions as per the processing capacity of MCC:

Processing Capacity of Bio-degradable waste in Tons Per Day	No. of Tubs	Size of Each Tub	Required Area for Processing Centre in Square Meter
5	14	3.9m x 1.8m x 1.0m	600
4	14	3.0m x 2.0m x 1.0m	410
3	14	3.0m x 1.5m x 1.0m	360
2	14	2.4m x 1.2m x 1.0m	260
1.5	14	2.0m x 1.2m x 1.0m	200
Source: (HUDD, Standard Operating Procedure for Decentralised Solid Waste Management in Odisha, 2019)			

10.4. Infrastructure and Machineries in MCC:

Table 10: Infrastructure and machineries at MCC

Sl. No.	Equipment and Infrastructure	Images	Specification	Quantity	Operational area required
1	Digital Weigh Bridge		Digital weigh bridge capacity up to 10 ton	1	12ft X 15ft
2	Belt Conveyor		Speed control from 5 ft/min to 30 ft/min, and 5Hp Motor-output – 500 kg/hr	1	10ft x 40ft
3	Shredder Machine		5Hp Motor, Capacity of Baling 50- 80 kg	1 for 3-5 TPD MRF	6ft X 6ft
4	Weighing scale		25-100 kg weighing capacity	1	3ft X 3ft

5	Sieving Machine/ Screening Machine		Sanitary Napkin-15-20 pieces per hour; Diaper- 1 to 2 per hour 900-degree min operation temperature	2 for 5 TPD MRF	5ft X 5ft
6	Fire Extinguisher		As recommended by Fire local authority		
7	Fabricated Trolley/ push carts			2-3	
8	Ancillary sorting tool: Fork tool, Shovel		As required		
9	Air tight drum (3 nos)				
10	Water Jet Machine				
11	Fly Trap				
12	Turbo Fan				

10.5. Type of Waste Processed at MCC:

All biodegradable waste as mentioned in the picture below from households and commercial units are processed at MCC.



Figure 3: Biodegradable waste that can be processed at MCC

10.6. Steps of Processing of Biodegradable Waste (Wet waste) at MCC:

The wet waste is processed at the Micro Composting Centre (MCC) to convert into organic compost by following a step-by-step process;

- Separating non-biodegradable waste from wet waste
- Preparation of Effective Micro Organism Solution (EM Solution)
- Preparation of tubs before loading waste
- Shredding of waste using a shredder
- Sequential loading of shredded wet waste in tubs and adding EM Solution
- Turning of waste and allowing aeration for decomposition of waste to convert into compost
- Removing compost from the tubs and drying
- Sieving the compost and packaging
- Quality check of compost



Figure 14: Process flow of wet waste composting; source: Urban Management Centre

10.6.1. Separating non- biodegradable waste from wet waste:

Before putting the wet waste in the tub, it has to be thoroughly checked and all the non-biodegradable wastes are removed from it.



After this, the waste is ready to be put inside the composting tubs.

Before placing the wet waste in the composting tubs, some prior preparations are required such as (a) preparation of Effective Microorganism (EM) solutions and (b) preparation of tubs as shown here in detail below:

10.6.2. Preparation and use of Effective Microorganism (EM) Solution:

EM stands for “Effective Micro-organisms”. EM consists of a wide variety of effective, beneficial and non-pathogenic microorganisms produced through a natural process and not chemically synthesized or genetically engineered. It is prepared as follows:



Figure 19: Process of preparing EM solution

- Take a barrel of 100 liters capacity.
- Add 3 liters of curd and 5 Kgs of jaggery. Mix it to form a homogeneous solution.

- c. Close the barrel and keep it fermented for 7 days.
- d. After 7 days, open the barrel and the EM Solution is ready to use.
- e. Take some EM solution in a jar and mix it with equal proportion of Rice bran and Rice husk.
- f. It should be used within three days of its preparation.

The quantity of EM solution should be such that the mixture can be turned into balls of the size of a coconut without breaking as shown in the figure-20. If it crumbles, add a little more EM solution to the mixture. If it is too much of watery, add more Rice bran and Rice husk.



Preparation of EM Solution



Preparation of EM solution balls

1. Keep the mixture ready to add on fresh waste.
2. The EM solution should be used within 3 days of preparation

Table 11: EM Solution as per MCC's utilization capacity

EM Solution as per waste quantity received at MCC						
	Waste received at MCC Tons Per Day (TPD)	Divide into 3 Barrels and Balls				
		 EM solution (Litre)	Use of EM Solution			
			 Day-1 (Litre)	 Day-2 (Litre)	 Day-3 (Litre)	 Make ball (Numbers)
In Tons	5	98	32.67	32.67	32.67	
	4	78	26.00	26.00	26.00	
	3	57	19.00	19.00	19.00	
	2	39	13.00	13.00	13.00	
	1.5	29	9.67	9.67	9.67	
Prepare "EM solution" in every 3 days interval keep it for 7 days for fermentation and used it for next 3 days after preparation						

EM solution is prepared in a sequence in every three days to keep the cycle continue and there is always ready-to-use EM Solution in stock at the MCC. The sequential preparation and use of EM Solution is shown in the following figure-21.

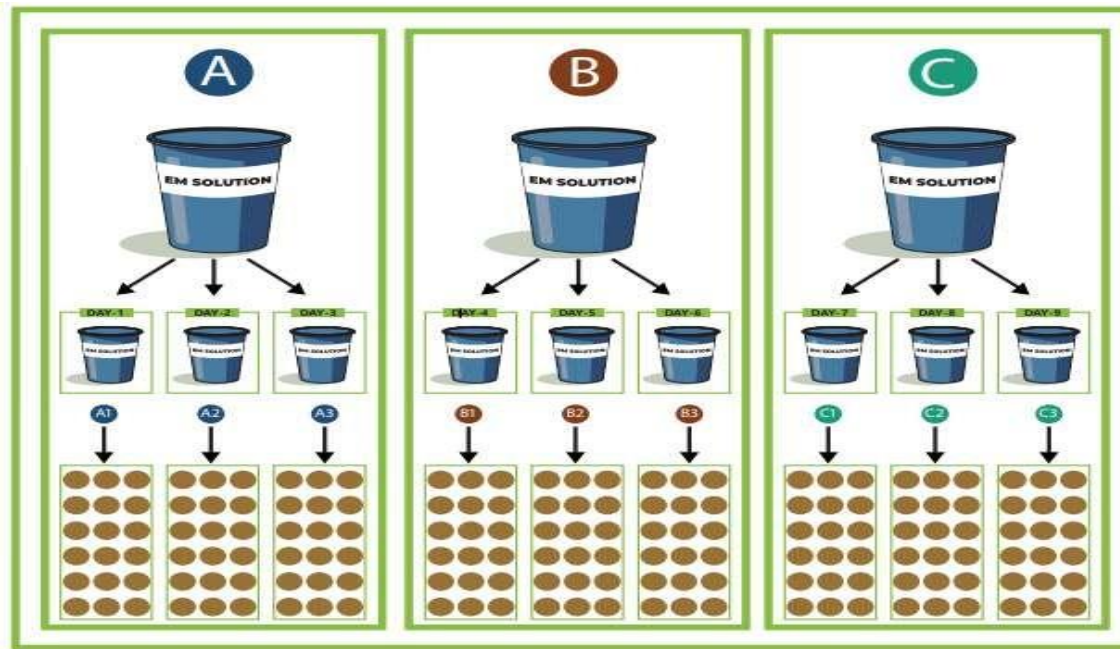
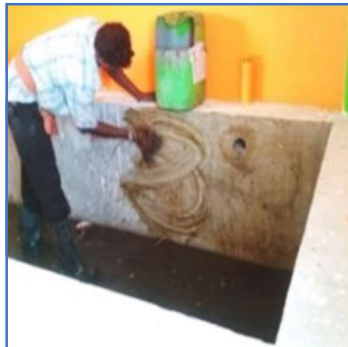


Figure 12: Preparation and use of EM Solution

10.6.3. Preparation of compost tubs to receive shredded waste for the first time:

- Put a 2-inch-thick dry cow dung at the floor of each tub, before putting waste in the compost tub for 1st time. This process is called bio-dozing.
- Then put the shredded biodegradable waste mixed with the rice-husk-mixed-effective microorganism-solution if not mixed during the shredding process
- Place the waste inside the tub and carry on the same process for next tub on next day.

Figure: Bio dozing of the tubs before loading waste



10.6.4. Loading of waste in tubs

Once the pre-required activities are done, the processing of wet waste is initiated.

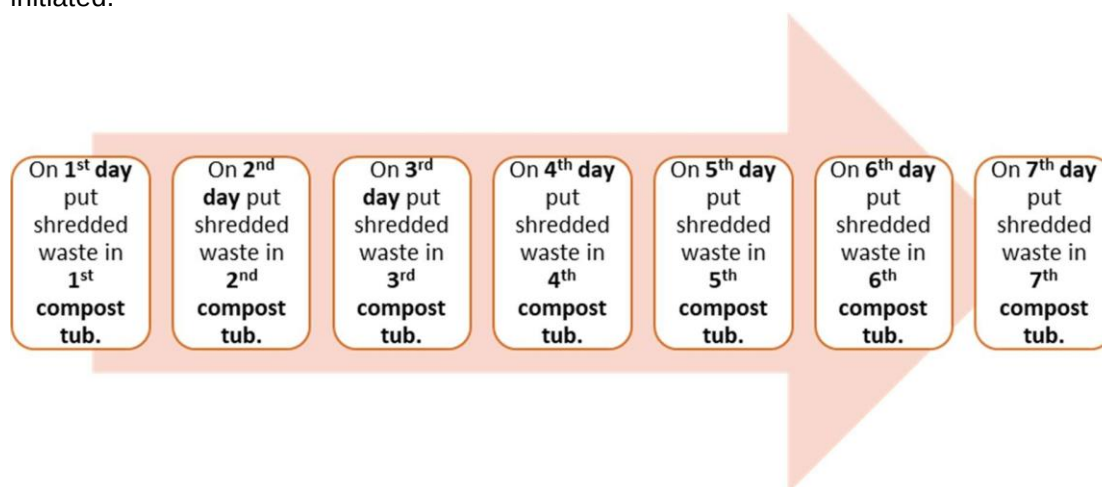


Figure 24: Sequence of putting waste in Tubs

- On 1st day, put the shredded waste mixed with prepared media in 1st tub.
- On 2nd day, put the shredded waste mixed with prepared media in 2nd tub.
- This method is to be continued till 7th day, on which the shredded waste mixed with prepared media will be put in 7th tub.
- The sequence of putting waste in tubs is: 1st day - 1st tub, 2nd day - 2nd tub, 3rd day - 3rd tub, 4th day - 4th tub, 5th day - 5th tub, 6th day - 6th tub, 7th day - 7th tub. For better understanding, the Figure-25 can be referred.

The same process has to be repeated every week, starting from day-1 to day-21 for putting waste in one train of tubs.

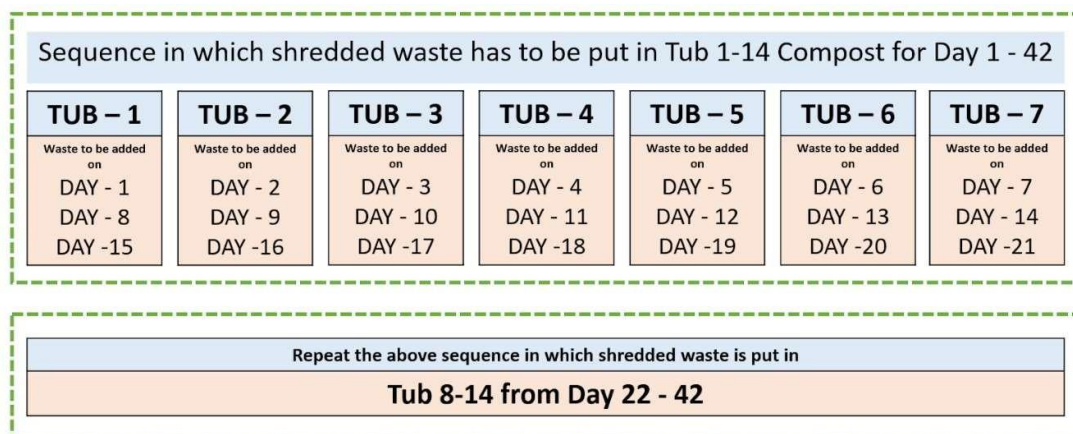


Figure 25: Sequence of putting waste in 7 tubs from Day 1 to Day 42

- e. Turn the waste up and down once in every two days for better aeration.
- f. With each passing day, the waste will degrade due to bacterial activity and there will be reduction in volume of waste. The colour of the waste will slowly turn brown and ultimately become black in final stage.
- g. After 42 days of adding waste in any tub, the waste in the tub would have turned black colour indicating matured compost.
- h. The matured compost shall be removed from the tub and sieved for uniform size.
- i. The rejects from the sieving shall be put again in the tub for further decomposition.
- j. The screened compost shall be kept for seven days for stabilization.

10.6.5. Turning of waste and aeration:

After sequential loading of wet waste in the tubs, turning of waste in every two days are key to aeration, which is important for decomposition of wet waste and quality of compost prepared. A detailed chart of loading of wet waste in the tubs and turning of waste in every 2nd day of loading waste for better aeration is provided below in the Table- 10, where green colour stands for loading of waste and yellow colour is for turning of waste.

It is important to ensure that 3/4 of the tub is filled with shredded wet waste, and 1/4 of the space remains vacant for better aeration. By turning the compost properly, the waste gets aerated and is mixed properly inside the tub, which will help foster aerobic conditions to break down the particles and fasten the composting process. Besides, good aeration within the tubs reduces foul odour in MCC.

Table 12: Loading and turning of waste in the tubs at MCC:

Day	Table-1						
	Tub-1	Tub-2	Tub-3	Tub-4	Tub-5	Tub-6	Tub-7
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
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41							
42							

10.7. Bio-degradable waste that cannot be processed in the tubs at MCC:

There are some biodegradable wastes which take long time for decomposition because of their composition, therefore, cannot be composted in the tubs of MCC. Some of these wastes are listed below for reference –

Not Easily Decomposable Waste



10.8. Management of Leachate:

Leachate is the juicy fluid that drains from wet waste. Leachate contains high value nutrients for decomposing. On the other hand, excess leachate slows down the decomposition process due to excess liquid content. Therefore, it is important to control the excess liquid by adding rice husk, sawdust, etc., which absorbs its liquid substance.

Besides, it is also important to allow the excess leachate drain through the strainer at the end of the tubs. Excess leachate passes through the leachate channel constructed between the tubs and connected to the leachate pit where it accumulates. For smooth draining of excess leachate, the tubs need to be constructed in such a way that the floor of each tub should be sloped to the side of leachate channel and have a large strainer fixed to the drain. The accumulated leachate will be rich in bacteria and can be used to moist the waste and accelerate decomposition. A small amount of leachate can be sprinkled on waste in tubs. However, care should be taken while doing so as more leachate creates an anaerobic condition in the tubs.

Leachate hole in the tub needs to be cleaned at least once in a week, so also the pipe connecting the hole to the Leachate Pit so that free flow of the Leachate from the Tub to the Pit takes place enhancing efficiency of the decomposing process.

Each tub needs to have sufficient number of holes on the side walls, which needs to be cleaned periodically to allow ventilation.

10.9. How to reduce foul odour, flies and maggots at MCC:

In general, MCC cannot be absolutely free from odours as waste if not fully decomposed would create foul odour. However, it is possible to control the odour if reason for the same is known. Primarily, foul odour is caused by the process of decomposition or breaking down of organic material. Reaction of bacteria on organic matter produces Hydrogen Sulphide (H₂S), the main cause of foul odour.

Table 13: In order to reduce the foul odour in MCC, the following actions are required:

Actions	Effect
Every 2 days, turn the waste in the composting tub	Internal odour will reduce
Add rice husk, sawdust to the waste in the tubs, if excess leachate is found	Reduces flies, maggots, and foul odour
Clean the leachate drainage line regularly (At least once in a week)	
Cover the leachate collection chamber properly with a lid which can be removed while cleaning	
Do not put fish or meat waste in the compost tub	Stops formation of maggots and flies
Install fans over the compost tubs and turbo fans on the roof to blow out air	Vents out air and odour
Install fly traps over each composting tub	Controls flies
Plant scented plants in the MCC premises	Reduces foul smells and creates a pleasant working environment

Do not leave post-processing waste lying around in the MCC for more than a day	Reduces foul odour and maggots
Clean the MCC premises, waste storage place, and waste carriers (cart, trolley, or tray) daily after waste is loaded in the tubs	Reduces foul smell
Do not leave used water in the open and properly clean the drains	Controls flies
Ensure free flow of air to the MCC as the basic principle of decomposition is “Aerobic”. The current structure of sheet dropping down at the end point of roof should be modified to allow free flow of air into the MCC. So also wall from plinth should be of minimum height to allow flow of air.	Facilitates decomposition and thereby odour controlled

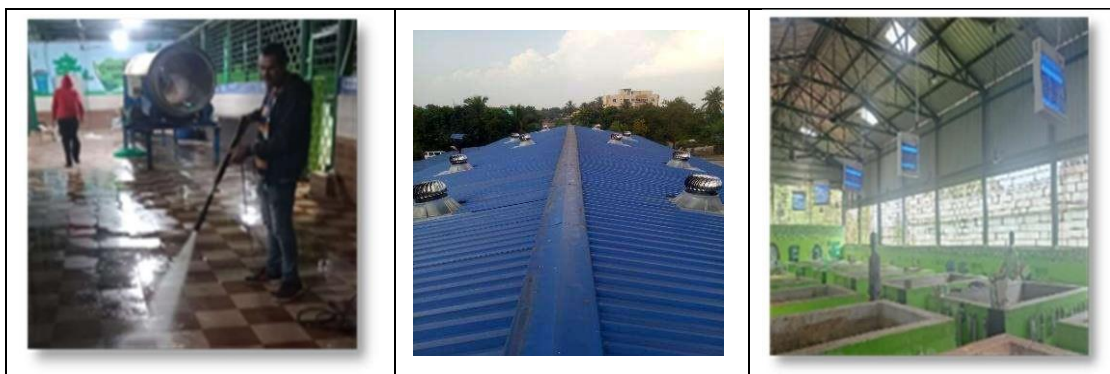


Figure 26: reduce the foul odour

10.10. Drying yard and Layout:

Every MCC must have at least two drying yards for spreading the compost after removing from the tub on the 42nd day for drying and stabilising.

A. Layout: The drying yard with concrete floor should be ideally of 50” X 10” (however, the ULB may decide depending on availability of space in the Wealth Centre) each with all the four sides open. The compost may still have moisture content when removed from the tubs. This makes it difficult for oxygen to penetrate the piles and can lead to unpleasant odours. So, drying yard must have saucer drain to allow the fluid drain out and these drains may be connected to the leachate chamber.

B. Roofing: Trusses with polycarbonate sheets for roofing may be used.

C. Leachate drain points must be connected to the drainage network for safe collection of leachates.

10.11. Compost Maturity:

The term “compost maturity” can be generally referred to the completion of the composting process. Maturity of compost can be determined through a simple test

known as “Stink Bag Test”. The compost is placed in a plastic bag and moistened to about 60% water content; the mouth of the bag is then closed (preferably sealed to prevent any airflow). The plastic bag containing the moist material is allowed to sit for 24-48 hours at room temperature (30°–35°C). If foul odour released when the bag is opened at the end of 24-48 hours, it indicates that the material is not fully matured and needs to undergo further decomposition. Generally, anything that smells bad is not good for crop or plant amendment.

The cured compost does not release odour because of carbon stabilization during aerobic decomposition of biodegradable material in tubs. Microbial activity continues during the curing phase also, but at a slower rate compared to the main composting phases.

10.12. Efficiency of wet waste processing:

Under ideal operational practices at MCC, the yield of compost can be around 20% of segregated processed municipal wet waste which also indicates good efficiency of the processing. The table-given below provides the monthly quantity of compost to be generated from wet waste at different level of efficiency rate of processing in MCC.

Table 14: Quantity of compost to be generated at different level of efficiency of processing in MCC (Tons per month)

	Capacity of MCC in Metric Tons Per Day	Capacity of MCC in Tons Per Month	Efficiency Rate		
			10%	15%	20%
In Tons	5	150	15	22.5	30
	4	120	12	18	24
	3	90	9	13.5	18
	2	60	6	9	12
	1.5	45	4.5	6.75	9

10.13. Quality of “Compost” Generated out of Wet Waste:

In order to ensure safe application of compost generated out of wet waste the specifications for its quality as indicated in Schedule-II of the Solid Waste Management Rules, 2016 shall be met.

10.13.1. Fertilizer Control Order:

The end product compost shall meet the standards prescribed under “**Fertilizer Control Order**” notified from time to time.

10.13.2. Features of Compost:

Compost is particularly useful as organic manure; it contains macronutrients (nitrogen, phosphorous, and potassium) as well as micronutrients (boron, iron, zinc, etc.). The use of compost reduces the dependency on chemical fertilisers for agricultural operations. When used as a soil amendment, compost reduces the need for water,

fertilisers, and pesticides. Compost acts as a soil conditioner, therefore supporting long-term fertility of the soil. The benefits of compost use include improved soil quality, enhanced water retention capacity of the soil, increased biological activity, micronutrient content, and improved pest resistance of crops. Also, consistent, high-quality compost is essential for building and sustaining the trust on the organic manure out of the Wet waste.

10.13.3. Concentration Limits:

Compost (Final product) exceeding the concentration limits stated under Schedule-II shall not be used for food crops. However, it may be utilized for purposes other than growing food crops. In other cases, it may be used for crops including food crops.

10.13.4. Testing of Compost:

Department of Fertilizers, Govt. of India has identified laboratories throughout India for testing of compost. Testing of the compost may be conducted in the designated Laboratory / Testing Centres at least once in a month.

Table 15: List of Government approved laboratories for compost quality testing in Odisha

Sl. No.	Address of the Laboratories
1	Analytical Chemist, State Fertiliser Quality Control Lab., Plot No. 646, Sahid Nagar, Bhubaneswar-751007 (Odisha)
2	Analytical Chemist, Fertiliser Quality Control Lab., Farm Road, P.O. Modipara, Sambalpur-768002 (Odisha)

The Department of Agriculture, Cooperation & Farmers Welfare, Govt. of India in consultation with State Governments has furnished (Letter No. F. No.11 026/08/2016-M&E, Dated 19th December 2019) the list of Soil Testing Laboratories identified by State Governments and the list of laboratories under the administrative control of the State Governments for use of testing of City Compost samples. The list of such laboratories in the state of Odisha is provided in below.

Table 16: List of Soil Testing Laboratories for Testing of Compost

Sl. No.	District	Address of Laboratory
1.	Angul	Soil Testing Laboratory, Near Krishi Bhawan, Angul – 759122
2.	Balangir	Soil Testing Laboratory, Agriculture Office Campus, Cold Storage Road, Balangir
3.	Balasore	Soil Testing Laboratory, At-Balia, Via Kuruda, Pin-758001
4.	Bargarh	Soil Testing Laboratory, Ekamra Chowk, At/Po-Sarsara, Bargarh, Pin-768040
5.	Bhadrak	Soil Testing Laboratory, Bye pass, Haldidiha, Bhadrak

6.	Boudh	Soil Testing Laboratory, DAO Office Campus, Boudh, Pin- 762014
7.	Cuttack	Soil Testing Laboratory, Jagatpur, Cuttack
8.	Deogarh	Soil Testing Laboratory, Near College Road, Deogarh
9.	Dhenkanal	Soil Testing Laboratory, Dhenkanal
10.	Gajapati	Soil Testing Laboratory, Po-Parlakhemundi, Dist-Gajapati
11.	Ganjam	Soil Testing Laboratory, Ankuli, Berhampur
12.	Jagatsinghpur	Soil Testing Laboratory, Jagatsinghpur, At-Nimakana, Po- Manijanga Block - Tirtol -754160
13.	Jajpur	Soil Testing Laboratory, At-Mansapola, O/o DDA, Jajpur
14.	Jharsuguda	Soil Testing Laboratory, O/o DDA Near SP Office, OMP line Jharsuguda
15.	Kalahandi	Soil Testing Laboratory, Near Arkabahali Farm, Bhawanipatna
16.	Mayurbhanj	Soil Testing Laboratory, Takhatpur, Baripada
17.	Kandhamal	Soil Testing Laboratory Agriculture Colony, Mandikunda Square, Phulbani
18.	Kendrapada	Soil Testing Laboratory, Kendrapada
19.	Keonjhar	Soil Testing Laboratory, DOA office campus, Madhapur, Keonjhar
20.	Koraput	Soil Testing Laboratory, Semiliguda, Dist.-Koraput Pin- 764036
21.	Khurda	State Quality Control Laboratory Building, Saheed Nagar, Bhubaneswar
22.	Malkangiri	Soil Testing Laboratory, Malkangiri
23.	Nuapada	Soil Testing Laboratory, Nuapada, Pin-766105
24.	Nabarangpur	Soil Testing Laboratory, DDA Office Campus, Miriganguda, Nabarangpur
25.	Nayagarh	Soil Testing Laboratory, Campus of Office of DDA Old Hostel Road, Nayagarh
26.	Puri	Soil Testing Laboratory, DAO Office campus, Chakratirtha Road, Puri
27.	Rayagada	Soil Testing Laboratory, Rayagada
28.	Sambalpur	Soil Testing Laboratory, Modipara, Sambalpur
29.	Sonepur	Soil Testing Laboratory, O/o DAO Campus, Near Royal Field, Sonepur

30.	Sundargarh	Soil Testing Laboratory. Rani Bagicha, Sundargarh
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10.14. Packaging and Marketing of Compost (“Mo Khata”):

In the Micro Composting Centres (MCC), after aerobic composting “wet waste” begets a new life and shape after being converted to the final product of “Compost” branded throughout the State as “Mo Khata”. This compost is the precious organic manure which needs to be made available to the customers (farmers) with pride and prudence. In the age of presentation and face value packaging is equally important as the quality of the product. Packaging should disseminate vital information to the customer, display the brand and quality guarantee and should look more attractive than the other brands of organic manure or chemical fertilizers.



Figure 28: Compost packaging in various sizes

The quality, texture, and even “look and feel” must be strategically designed. A branding of the “Mo Khata” should be consciously decided to popularize the product. The compost should be packed in bags of **various sizes** and it should be made up of **suitable materials**. The packaging should be such that it is **handy to carry** and has a clear and attractive printing to attract customers. The **basic information** regarding the product should be displayed on the bag appropriately. The bags should be **sealed** properly using sealing machine to maintain the quality of product. Overall, the packaging should be informative to the customer, display a reliable brand and quality guarantee, and look at least as compelling as that of competing soil amendments.

10.15. Promoting Compost:

For marketing purposes all the Line Departments concerned having programme of plantation should be kept in touch for obtaining indent well ahead of the commencement of plantation season from them for timely supply of compost which may be delivered at the designated place by own arrangement by the ULBs.

Nearby Blocks and other offices may be contacted for reaching out to various stakeholders under their administrative set up. Agriculture / Horticulture authorities may be consulted for reaching out to the farmers.

IEC activities may be carried out to popularise “Mo Khata” amongst people.

The In-Charge of the Wealth Centre shall enter the relevant entries on production and sale of the compost on daily basis in the Mobile App “AMA SAHARA” and verify the stock of the day.

10.16. Managing inventory of “Mo Khata”:

The inventory of “Mo Khata”, the compost generated from the processing of wet waste at MCC, is maintained in a register as well as recorded in the Ama Sahara App. While recording the stock of Mo Khata, only ready-to-sale stock is to be entered, not the compost kept in the drying yard for stabilising.

10.17. Expenditure out of resources generated from sale of “Mo Khata”:

The “Mo Khata” generated in the MCC shall be the deemed property of the ULB. The resources generated from the sale of “Mo Khata” as per the price fixed by Dept./Govt. from time to time shall be deposited in the Corpus Fund of the Wealth Centre and shall be utilized to meet operation and maintenance cost of the “Wealth Centre”, payment of monthly incentive to the Community Partners, cost of transportation of “Mo Khata” for delivery to the Line Departments and matters ancillary and incidental with maintenance of records on all such expenditures. Money Receipts should be issued for receipt of the amount towards sale of “Mo Khata” and the data should be entered in “Ama Sahara App”.

Use/ sale of “Mo Khata”	
Complimentary supply to households	To realize the importance of source segregation, to deliver the message that the “Waste is Wealth”, and to replace a chemical fertilizer with “Mo Khata”, “Mo Khata” should be supplied to the households in the service area as per request. Mo khata is to be provided at 20% discounted rate to the Swachha Sathi/ Swachha Karmi (Swachha Sabari Driver and Helper for door-to-door sale of Mo Khata to households and commercial units and bulk generators including institutions and offices
Use in parks	ULB should use “Mo Khata” in the plantation made in places under its administrative control such as parks, plantations made in the road dividers and other places. But it has to pay the required amount for lifting “Mo Khata”.
Memento	It can be presented as mementos in different competitions, awareness campaign, felicitation, Orientation meetings etc. as it carries a significant message on Environmental Protection, care for the waste, and in promoting the use of organic compost. Behavioural change in exponential form can take place through this activity. However, cost towards the amount of “Mo Khata” taken for use as Memento should be deposited with the “Wealth

Use/ sale of “Mo Khata”	
	Centre”. The district administration and other Govt & private organizations may be impressed upon to use “Mo Khata” as a memento in their functions & events to promote the cause of environmental protection.
Outlets for sale	ULBs may through the members of the SHGs/ Transgender Groups or on their own open outlets in suitable places and make available “Mo Khata” for purchase by people at the aforesaid price.
Online sale	ULBs may also take up in a professional approach to enter the market, to make available their proud product to people in the highest level of circulation. On-line marketing being one such tool, they may take up the needful action to popularize the product not only inside Odisha but join in a Pan India movement.
Sale through GeM portal	Currently, compost is not sold in Government e Market (GeM) Portal. Therefore, ULBs may explore scope to sell compost via this transparent channel of marketing through GeM Portal.
Supply to line Departments	Departmental Authorities of the Agriculture & Farmers’ Empowerment Department, Forest Environment & Climate Change Department etc. may be supplied with the compost at the aforesaid price on receipt of the indent. Proactive steps be taken for supply to the Line Departments in time.
Request by households	Request for door delivery of “Mo Khata” made by the households through the Swachha Sathis under her coverage area may be accepted and small quantities may be supplied through the Battery-operated Vehicles for door delivery, in case of larger requests, special BoV may be deployed engaging the Members of the SHGs/Transgender groups assigned with operation of the “Wealth Centres”. Incentive as indicated above may be paid to the Community Partners (Individual members / Group as the case may be).
Bulk buyer	The Bulk Purchaser (Purchase of 100 Kg. or more in one go) may be issued with a Letter of Appreciation by the ULB for promoting the use of organic manure and thereby aiding in Environmental Protection and sustenance of the Wealth Centre side by side.
Farmers	Farmers may be the target Group for sale of compost and to promote use of compost so as to fulfil the obligation to phase out the use of chemical fertilizer.
Other modalities/ strategy	Any other modalities / strategy found suitable for sale / disposal of “Mo Khata” may be taken up to make available “Mo Khata” to the persons/ institutions / Department requesting for the same.

11. Performance Indicators for MCC:

The performance level of the MCC is to be assessed by examining / evaluating on the basis of the following criteria:

- a. Percentage of days the compost plant was not operational
- b. Quantity of waste received
- c. Compost production (kg)
- d. Pre-processing reject in percentage
- e. Post-processing rejects in percentage
- f. Compost yield (with an acceptable limit)
- g. Percentage of Reject (with an acceptable limit)
- h. Capacity utilization of the compost plant
- i. Quantity of compost sold
- j. Revenue generated from selling of compost
- k. Expenditure on the operations
- l. Expenditure on maintenance
- m. Shredder downtime (in days) in percentage
- n. Sieving machine downtime (in days) in percentage
- o. Flytrap downtime (in days) in percentage
- p. Total stock of compost
- q. Percentage of absenteeism of human resources
- r. Monthly Testing of compost in approved Laboratory

12. Material Recovery Facilities (MRF)

12.1. Conceptual design:

As per the SWM Rules 2016, a MRF is defined as a facility where non-compostable solid waste can be temporarily stored by the local body or any other entity or any person or agency authorised by any of them to facilitate segregation, sorting and recovery of recyclables from various components of waste by authorised informal sector of waste pickers, informal recyclers or any other work force engaged by the local body or entity mentioned in rule 3 for the purpose before the waste is delivered or taken up for its processing or disposal.

The Material Recovery Facility (MRF) is an essential part of a “zero waste” management programme that reduces dependence of landfill as it recovers recyclables from the discarded materials and adds value to those for marketing to end-user manufacturers. The main function of the MRF is to maximize the quantity of recyclables recovered from collected waste that will generate the highest possible revenue in the market.

12.2. Evaluation of the market and economics of operation:

It is important to devise a recycling strategy whereby the waste management value chain is functional and no residual waste stream is left behind and. The primary objective of MRF is to sort, store and pre-process dry waste like baling before sale to buyers/recyclers. Since waste streams vary in composition and volume from one place to another, the MRF should be designed specifically to meet the waste management goals of the city and its adjoining rural areas in the long term. Odisha aims at a seamless convergence of rural and urban waste management. Therefore, the cities should do a careful assessment the quantity of non-bio degradable waste generated in the city as well as in the rural areas tagged to it and accordingly plan the MRF.

On the other hand, it is important to map the recyclers' market and tie up with them to maximise the uptake of the recyclable materials at a competitive price. The ULBs need to short list and empanel the recycler/ buyer of recyclables with item-wise rate by adhering to the prescribed process.

12.3. Infrastructure Planning for MRF:

The land and human resources requirement for establishment of MRF depend upon the quantity of dry waste to be processed. According to advisory on MRF published by CPHEEO in 2020, an MRF of capacity of 5 TPD would require an area of 1500-2500 sqm.

Based on the learning from the existing MRFs in Odisha, it is calculated that the average area required per tonne is about 160 sqm per tonne (i.e. 0.06 tonnes processed per sqm). Also, the per capita dry waste generation in the State is estimated to be 150 g per day.

ULBs can use these conversion rates to optimize the available space and estimate the population that can be served by building MRF in a given area. The detail calculation is provided in Table below.

Additional MRFs should be constructed to meet the demand and have adequate facility to process the dry waste with ease.

Table 17: Sample Calculation of area for processing waste

- | |
|--|
| <ul style="list-style-type: none">▪ Space available: 1000 sqm▪ Waste processed per sqm: 0.006 tonnes or 62.5 kg▪ Total waste processed: $1000 \times 0.006 = 6.25$ tonnes▪ Population served per tonne: 1,50,000▪ Total population covered: $6.25 \times 150000 = 937500$▪ Average size of households: 4.5 people▪ Total household covered: $937500 / 4.5 = 208333$ |
|--|

Also, based on empirical evidence, two human resources are required per tonne to process dry waste at the MRF. A one tonne processing capacity MRF can process the waste of 1.5 lakh population, which covers 33 thousand households. The above

estimate can be used in estimating the additional area and human resources required for expansion of an existing MRF.

The Table below provides a detailed estimate of area and human resources required per tonne of waste received at MRF per day. This includes basic infrastructure for sorting shed, storage space for sorted recyclables and other basic infrastructure such as record room and water and sanitation facility, etc. (CPHEEO & MoHUA, 2020).

Table 18: Estimate of area and human resources for MRF

Sl. No.	Capacity of MRF (TPD)	Area Required (sqft) @160 per tonne	HR required @1.6 per tonne	Population	Household covered
1	1	160	2	150000	33333
2	2	320	3	300000	66667
3	3	480	5	450000	100000
4	4	640	6	600000	133333
5	5	800	8	750000	166667
6	6	960	10	900000	200000
7	7	1120	11	1050000	233333
8	8	1280	13	1200000	266667
9	9	1440	14	1350000	300000
10	10	1600	16	1500000	333333

The above table considers a single shift. The second shift, if needed, can be decided with 40% of the human resources as primary sorting would be completed in the first shift itself. These numbers are indicative and may vary based on local scenario.

12.4. Type of Municipal Solid Wastes managed in MRF:

12.5. Non-Biodegradable wastes/ Dry wastes:

As mentioned earlier, MRF manages the “dry waste” means waste other than biodegradable waste and inert street sweepings such as and includes recyclable and non-recyclable waste, combustible waste and sanitary napkin and diapers, etc.

Under the dry waste/ non-biodegradable waste, several other waste types find their way into the municipal solid waste stream, which require special handling and disposal because of their quantity, concentration, physical and chemical characteristics, or biological properties (CPHEEO & MoUD, Municipal Solid Waste Management Manual - Part 2, 2016) which are also managed at MRF:

Sanitary Waste:

Wastes comprising of used diapers, sanitary towels or napkins, tampons, condoms and any other similar waste (CPHEEO & MoUD, Municipal Solid Waste Management Manual - Part 2, 2016). The waste generators are required to handover sanitary waste safely wrapped in pouches to the waste collector of the local body. Upon collection of

sanitary waste, it should be preferably disposed of in biomedical or MSW incinerators at the MRF, as applicable to the local context or as directed by the SPCB as per the standards mentioned in the SWM rules, 2016.

Domestic Hazardous Waste:

As per the SWM Rules, 2016 the “domestic hazardous waste” means discarded paint drums, pesticide cans, CFL bulbs, tube lights, expired medicines, broken mercury thermometers, used batteries, used needles and syringes and contaminated gauge, etc., generated at the household level. It is local bodies responsibility to safely handle and dispose of the domestic hazardous waste under the directions from SPCB through the MRF.

Plastic Waste:

The “plastic waste” means any plastic product such as carry bags, pouches or multi-layered packages which have been discarded after use or after their intended life is over. (CPHEEO & MoUD, Municipal Solid Waste Management Manual - Part 2, 2016). Plastic waste is managed at MRF - segregation, collection, storage, transportation, processing and disposal through authorized Plastic Waste Processors (PWPs) - as mentioned in the rule 6(1) of the Plastic Waste Management (Amendment) Rules, 2018.

E-Waste:

Means electrical and electronic equipment, whole or in part or rejects from their manufacturing and repair process, which are intended to be discarded as waste The waste generators will hand over the E-waste to the waste collector of ULB. The e-waste management rules specify that the collected e-waste at the MRF should channelize the e-waste to recyclers and play the role of Extended Producer Responsibility (EPR).

12.6. Equipment and Infrastructure at MRF:

The basic equipment/machineries and furniture used in a semi-automated MRF are provided in the Table -19 below.

Tabl2 19: List of basic equipment/machineries and furniture used in a semi-automated MRF

Sl. No.	Equipment and Infrastructure	Images	Specification	Quantity	Operational area required
1	Digital Weigh Bridge		Digital weigh bridge capacity up to 10 ton	1	12ft X 15ft

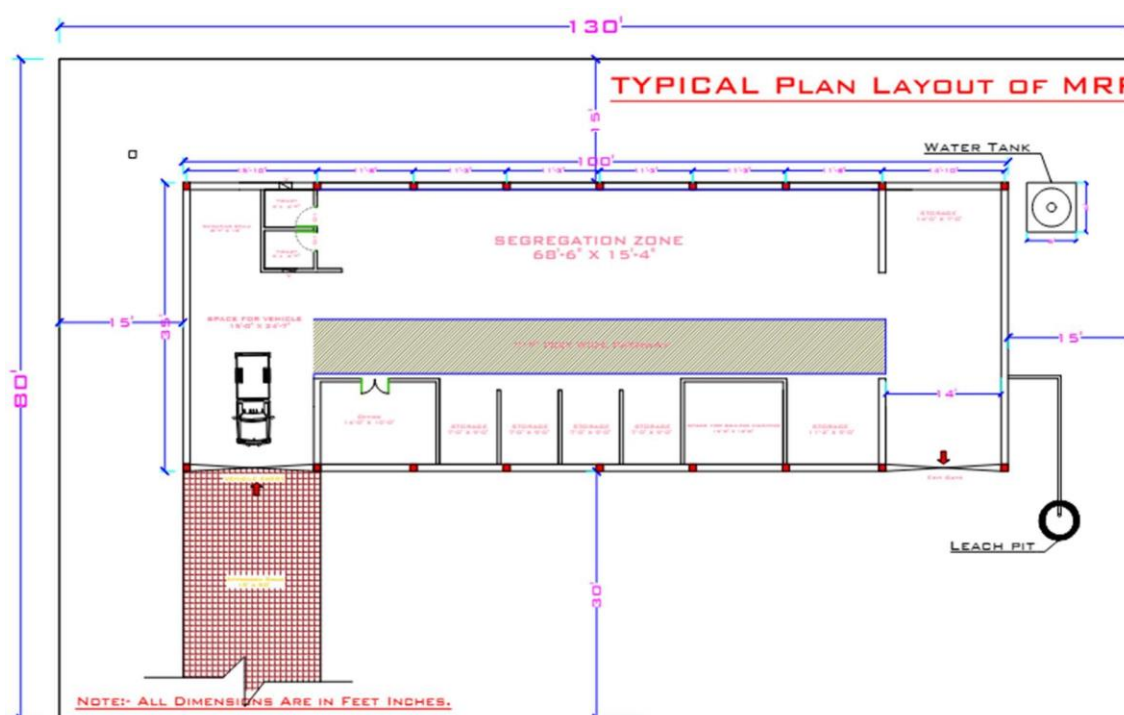
2	Horizontal Belt Conveyor		Speed control from 5 ft/min to 30 ft/min, and 5Hp Motor-output – 500 kg/hr	1	10ft x 40ft
3	Baling machine		5Hp Motor, Capacity of Baling 50- 80 kg	1 for 3-5 TPD MRF	6ft X 6ft
4	Weighing scale		25-100 kg weighing capacity	1	3ft X 3ft
5	Incinerator		Sanitary Napkin-15-20 pieces per hour; Diaper- 1 to 2 per hour 900-degree min operation temperature	2 for 5 TPD MRF	5ft X 5ft
6	Fire Extinguisher		As recommended by Fire local authority		
7	Fabricated Trolley/ push carts			2-3	
8	Collection Bins: for sorted recyclable wastes and for inert waste		Large bins size 3ft x 2.5ft		
9	Ancillary sorting tool: Fork tool		As required		

10	Furniture Cane/ Plastic Stools		Height 30 inch		
11	Aluminium Ladder		6-8ft height		

12.7. Facility Design and Layout of MRF:

The facility design plan of MRF should accommodate all the equipment discussed above and their operation along with all the activities which promote efficient and effective operation of the MRF. The facility design and layout are furnished below in Figure.

Figure 31: Layout Design of MRF



12.8. Operation process of MRF:

The operation process flow of an MRF in Odisha has broadly six steps:

1. Preliminary segregation of sortable dry waste for loading on conveyor
2. Detailed sorting of dry waste into various categories on conveyor
3. Baling and bundling of Dry Waste

4. Storage of dry waste
5. Sale of dry waste to empanelled sellers/recyclers
6. Consignment of non-recyclable fraction of dry waste to cement companies for co-processing

Table below provides a broad list of materials; their type and recyclability are provided as a reference for the ULBs.

The MRF should have weighbridge and conveyor system to bring in efficacy to the system, Funds placed for the purpose may be utilised immediately for placing these in place.

Table 20: Classification of Materials as per type and recyclability

Material Classes	Material Types	Recyclability
Recyclable Paper	Cardboard	Recyclable
	Newspaper	Recyclable
	Cartons	Recyclable
	Mixed Paper	Recyclable
Recyclable Glass	Glass Containers	Recyclable
Recyclable Metal	Aluminium Cans	Recyclable
	Aluminium Foil and Trays	Potentially Recyclable
	Tin Food Cans	Recyclable
	Empty Aerosol Cans	Potentially Recyclable
Recyclable Plastic	PET (#1) Bottles and Jars	Recyclable
	PET (#1) Small Rigid Plastics	Potentially Recyclable
	Clear HDPE Bottles and Jars	Recyclable
	Coloured HDPE (#2) Bottles and Jars	Recyclable
	HDPE (#2) Other Containers	Potentially Recyclable
	LDPE (#4)	Potentially Recyclable
	PP (#5) Bottles and Jars	Potentially Recyclable
	PP (#5) small Other Rigid Plastics	Potentially Recyclable
	PS (#6) Rigid Plastics	Potentially Recyclable
Foam Plastic	EPS Food Packaging	Potentially Recyclable
	EPS Foam Blocks and Shapes	Potentially Recyclable
Film Plastic	Clean Plastic Bags and Film	Potentially Recyclable
	Disposal Bags	Potentially Recyclable
	Other Plastic Film	Potentially Recyclable
Other materials	Household Hazardous Waste	hazardous- needs safe disposal
	Electronics and Small Appliances	Recoverable
	Diapers and sanitary napkins	Contaminates – needs safe disposal by incineration
	Textiles/ Shoes	Non-recyclable – needs safe disposal
	Furniture	Recoverable

	Mixed Residue	Contaminates – needs safe disposal
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12.8.1. Preliminary Segregation of Sortable Dry Waste:

- The dry waste received at MRF is thoroughly checked for residual wet/mixed waste at the primary sorting area.
- The wet waste is removed and transported to MCC.

12.8.2. Detailed sorting of dry waste on a conveyor belt:

The detailed sorting of dry waste will be as per the following steps:

- The dry waste is taken to a horizontal conveyor belt using a trolley or push cart.
- The dry waste is placed on the horizontal conveyor belt to sort the dry waste into the different categories based on details provided in Table below –

Table 21: Categories of dry waste for sorting at MRF

Sl. No.	Category of dry waste	Type	Examples of Waste
1	Plastic	Polyethylene Terephthalate (PET)	Jars, Bottles, Ropes, Carpet, Tote Bags, Combs
		High-Density Polyethylene (HDPE)	Containers, Grocery Bags, Shampoo Bottles, Toys, Trash Bags
		Polyvinyl Chloride (PVC)	Plumbing Pipes, Tiles, Shoes, Ducts, Sewage Pipes
		Low-Density Polyethylene (LDPE)	Cling Wrap, Frozen Food Wrapping, Condiments
		Polypropylene (PP)	Tupperware, Kitchenware, Plastic Diapers, Disposable plates and cups, bottle caps
		Polystyrene (PS)	Disposable coffee cups, Plastic Cutlery, Peanut packaging
		OTHERS/Multi-Layer Plastics (MLPs)	CD's, DVD's, Eyeglasses, bottles, Tetra packs, Food Packaging (Potato Chips Packets)
2	Paper	Corrugated Cardboard	Packaging boxes, Egg Cartons, Shoe boxes, cereal boxes
		Mixed Paper	Mail, Catalogues, Phonebook's magazines
		Newspapers (ONP)	Newspapers
		High Grade Inked Paper	Envelopes, Copy Paper, Letter Heads
3	Metals	Aluminium	Aerated Beverage Cans, Aluminium Foils
		Steel	Food Cans, Stainless steel mugs, Toys, Lids

		Tin	Cans, Household containers
		Iron	Cooking Instruments
4	Wood	Wood	Broken Furniture, Blocks, Boxes
5	Glass	Container Glass	Bottles, Jars, Glassware, Bulbs
6	Leather, Rags and Rubber	L and R	Garments, Shoes, Bands, Wires
7	Thermocol	Polystyrene Foam	Styrofoam, Small or Large Packaging

The use of conveyor belt enhances the sorting efficiency by three times compared to conventional method of sorting. Besides, it also improves the posture and convenience of Swachha Karmis making the process easier and comfortable.

12.8.3. Baling and Bundling:

The baling and bundling of dry waste comprise of the following steps -

- High volume of plastic waste/PET bottles, MLPs, cardboard etc. is baled into rectangular bales and bound and stack in the storage area.
- Other materials are similarly packed in bags or bundled and stored in the storage area
- Lightweight materials can be stacked on top of each other to optimize space.

Figure 32: Incinerator at MRF



Figure 33: Baling of sorted waste

Sanitary Waste and domestic hazardous wastes like used diapers, sanitary towels or napkins, tampons, condoms, incontinence sheets, masks and any other similar waste is separated from the dry waste and incinerated in the napkin incinerator in MRF till the waste is reduced to ash.

The ash residue after incinerator is collected and stored in a specific container for onward safe disposal.

12.8.4. Cleaning and upkeep of MRF:

The daily operation of MRF will be closed with sweeping of the floor of the primary sorting area and cleaning with water by using a jet spray if it has residue of wet waste or mixed waste.

All MRFs should keep the primary sorting area clean to receive fresh waste next day. No MRF should have mixed waste dumped at the primary sorting area.

12.9. Marketing and sale of recovered materials and safe disposal of non-recyclables:

12.9.1. Downstream Processing and Compliance

After segregation of dry waste in multiple categories, the waste is sold to either scrap dealers or recyclers. The dry waste can be categorised into three waste streams based on downstream processing.

The Table below gives details of type of downstream processor and statutory required for different types of dry waste

Table 22: Downstream process and compliances for different types of waste

Sl.No.	Type of Waste	Type of Processor	Compliance Requirement
1	Paper	Reseller/Paper Mills	Relevant Compliances
2	Recyclable Plastic Waste (PET, PVC, HDPE, LDPE)	Reseller/Plastic Waste Recycler	CTE and CTO required for resellers if operations are more than 5 TPD Authorization under Plastic Waste Management Rules 2016 required for Plastic Waste Recyclers
3	Metal	Reseller/Foundries	Relevant compliances
4	Non-Recyclable Waste	Cement Companies	Authorization from SPCB, Odisha
5	E-Waste	Recyclers/Dismantlers	SPCB, Odisha Authorization
6	Domestic Hazardous Waste	Hazardous Waste Recyclers/TSDF	SPCB, Odisha Authorization

12.9.2. Empanelment of buyers/recycler:

The ULB can either sell the waste directly to Recyclers/Processor authorized by SPCB (under the Plastic Waste Management Rules 2016), if they have sufficient quantities or an intermediary reseller, who will then aggregate waste from other sources and sell the waste to a recycler.

The e-waste and domestic hazardous waste have to be sold to an authorized recycler under e-waste management rules. The ULBs can empanel the reseller/recycler through a tendering process. The MRF can also store the e-waste/domestic hazardous in case the empaneled recyclers require a minimum quantity for picking up the material.

12.9.3. Procedure of selling to recyclers:

The ULBs should follow the following steps for empanelment of vendors.

- Set up a benchmark for selection of vendors
- Draft an RFP for the empanelment of vendor
- Invite applications for interested parties for sale of waste to RFP
- Onboard agencies for sale based on the benchmarks and cost
- Maintain sale register to capture sale details
- Enter sale details in Ama Sahara App.

The RFP should contain the following points:

- Details of statutory requirement as per the SWM Rules, 2016 and
- Details of downstream partners
- Indemnification of the ULBs
- Rate Contract and frequency of collection

In addition to this, officials of the ULBs should visit the facilities of reseller to ensure that no malpractices like child labour and use of potentially hazardous techniques for processing of waste is followed by the resellers/recyclers, which is violating the existing norms.

12.9.4. Safe Disposal of non-recyclables:

The non-recyclable materials may be stored, baled and sent to cement factories for co-processing. H&UD Department through Letter No. 11387 Dated 01.07.2019 has partnered with 4 cement factories in Odisha for safe disposal of non-recyclable materials. H&UD Department has designated nodal ULBs located in the proximity of these cement factories, which can aggregate waste from other ULBs and sent the cumulative waste to cement factories.



The Table below gives an indication for cement plants who can co-process non-recyclable dry waste at the facility –

Table 23: List of Empanelled Cement Factories in Odisha

Sl. No.	Name of Empanelled Cement Factory	Nodal ULB
1	ACC Ltd.-Cement Works, Cement Nagar, Bardol, Bargarh	Bargarh
2	OCL India Ltd., Rajgangpur, Sundargarh	Rajgangpur
3	Shiva Cement Ltd., Unit – II, Telighana, Kutra, Sundargarh	Rajgangpur
4	Toshali Cement (P) Ltd., Ampavalli, Koraput	Koraput

13. Managing Sales Proceed:

The proceeds from the sale of materials will be deposited in the Corpus Fund of the Wealth Centre. The revenue so generated may be utilised in the manner as indicated in the preceding paragraphs.

14. Performance Indicators for MRF:

The performance level of the MRF may be assessed by examining / evaluating on the basis of the following criteria:

- a. Quantity of waste sent for co-processing
- b. Quantity of rejects in percentage
- c. Quantity of recyclables sold
- d. Revenue generated from the selling of recyclable waste
- e. Expenditure on the operations
- f. Expenditure on maintenance
- g. Baling machine downtime (in days) in percentage
- h. Napkin incinerator downtime (in days) in percentage
- i. Percentage of absenteeism of manpower
- j. Monthly testing of compost from laboratory
- k. Percentage of days the MRF was not operational
- l. Quantity of waste received
- m. Quantity of waste recycled

Periodic assessment of performance may be made to resort to curative measures and improve quality of service. The State Level Monitoring Cell may be intimated of the outcome of the review.

The Cell suo-motu shall also assess with the help of the Mobile App AMA SAHARA and constant monitoring the smooth functioning of the Wealth Centers.

The provisions should be scrupulously followed for implementation for enhancing efficacy of the Wealth Centre and strengthening the Sanitation Value Chain.

15. Maintenance of equipment at Wealth Centre:

Maintenance of wealth centres includes repair and replacement of infrastructure, machineries, equipment, fixtures (plumbing, sanitary and electrical) and any other repairs wherever. Maintenance can be categorised into two types:

- a. **Preventive maintenance** is to check or inspect regularly to lessen the likelihood of sudden break down. This includes checking or inspecting of equipment, machinery, electrical fixtures such as switchboard, electric motor, pump, etc. and sanitary fixtures such as taps and drainpipes.

b. **Responsive maintenance** takes care of remedial action to resolve any breakdown. ULB shall engage appropriate agencies/ITIs for annual maintenance of infrastructure, machinery, and vehicles through a contract.

Table 24: List of required maintenance tasks for the Wealth Centre

List of required annual maintenance in Wealth Centre		
Task (in-charge)	Category	Frequency
Equipment/Machineries		
Washing and cleaning of waste collection vehicles (BoV) by Swachha Karmis/ Driver & Helper.	Preventive	At least twice a week
Inspect all machineries (Weigh Bridge/ Weighing Scale, Baling Machine, Incinerator, Conveyor belts and sieving machine) and hire agency for repair/ replacement through annual maintenance contract (AMC). Periodic inspection by agency under AMC.	Preventive	Once a month
Coordinate with agency under AMC when there is damaged or dis-function of machineries and equipment.	Responsive	As and when required
Electrical Fixtures		
Inspect all bulbs, tube lights, fans, hand dryers, switches, plugs, miniature circuit breaker (MCB) switch and any other electrical fittings in the wealth centre and repair/ replace as needed	Preventive	Once a month
Repair and replace electrical fixtures when these are damaged or not functioning	Responsive	As and when required
Plumbing Fixtures		
Inspect all the toilet seats, urinals, wash basins, flush knobs, taps, faucets, showers, pipes, and any other plumbing fixtures within the wealth centre including leachate channel to identify any leakage/ damage/ malfunction that may need repair/ replace	Preventive	Once a month
Other elements		
Inspect sanitary pad vending machines and restock as needed. Inspect sanitary pad incinerators and clear out the ash as needed as per the instructions of the manufacturer.	Preventive	Once a month
General Infrastructure		
Report any visible damage in the internal/external walls/ceiling of wealth centre building, large cracks and breakages, damage in leachate tank/soak pits/septic tanks and damage in the drains	Responsive	As and when required
Inspect all the doors, windows, ventilators, latches, hooks, and any other elements within the wealth centre for any damage/ breakage that may need repair/replace	Preventive	Once a month
Maintain a barrier free access to the wealth centre by not allowing any other activities at or close to the entrance of the wealth centre	Preventive/ Responsive	As and when required

Inspect to identify any damage in painted walls and signage, and inform the ULB in case of requirement of renovation/replacement	Preventive	As and when required
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A maintenance register is to be maintained at the Wealth Centre to capture details of breakdown and maintenance activities at the Wealth Centres.

16. ISO Certification:

Third Party assessment may be made for ascertaining the qualitative maintenance of the infrastructures for which ISO Certification for each MCC and MRF be made, Funds from the Corpus Fund / sale proceeds available from sale of “Mo Khata” or Recyclables may be utilised for the purpose.

17. Information Education and Communication (IEC):

17.1. Display boards:

All wealth centres shall have standardised display of information (prototype is provided in Annexure-C). The following display boards are mandatory for every MRF:

- a. Board 1: DOs and DON'Ts at Wealth Centre
- b. Board -2: Daily Update on Waste
- c. Board -3: Process flow of MRF
- d. Board -4: Entitlement of Swachha Karmi
- e. Boards-5: Safety Measures at Wealth Centre

17.2. Awareness Campaign:

The ‘Source Segregation’ being the most important parameter across the solid waste management value chain, it is necessary to communicate the message at household level through IEC and BCC campaign. For this there is a need to develop inter-personal communication to generate awareness on source segregation and its importance in managing the solid waste.

The Swachha Sathis plays a very important role to sensitize all the units assigned to them.

- Sensitize public and educational institutions through practical demonstration on waste segregation at source.
- Take up vigorous actions to orient the waste generators in practicing source segregation in day-to-day life.
- Aware waste generators on the rate of user fees to be paid by them and the rebate available for payment of the User Fee in one go for a year by the Swachha Sathis.
- Identify houses who provides segregated waste and tag them as Swachh Gruh
- Identify houses who does not provides segregated waste and tag them as Aswachh Gruh.

The ULBs shall also take appropriate awareness campaigns based on the following key messages as provided on Table below:

Table-25: Key messages for IEC campaign

Sl. No.	Key Messages
1	Behavioural changes of waste generators for keeping dustbins and handing over segregated waste to collection vehicles of ULB
2	My waste my responsibility – Green bin for wet waste and blue bin for dry waste
3	Refuse, reduce, reuse, recycle, recovery of waste – everyone's responsibility
4	Pay a nominal user fee through Ama Sahara App by waste generator for collection of waste from door step by ULB
5	Use ULB's door-to- door waste collection facility and be a part of the state's mission of waste management

Apart from educating the waste generators, IEC and BCC for Swachha Karmis is also required for various activities at the wealth centre.

- Dissemination of educational messages for awareness generation of various stakeholders engaged should be done.
- Displaying the educational messages about their responsibilities at their respective work areas.
- For example, the responsibilities of a Swaccha Karmi at MCC for processing the wet waste can be displayed by painting it on the outer walls of composting tubs, etc.
- Colourful wall paintings related to sanitation and solid waste management may be displayed suitably on the walls of the Wealth Centres to disseminate messages on environmental impact of scientific processing of solid waste and other related matters.

It is the primary responsibility of ULB to generate awareness among the staff of the wealth centre about the benefits of organic compost. Following activities can be taken up by the ULB for promoting sale of the organic composts generated at the wealth centre:

- The farmers and other stakeholders of surrounding area should also be educated upon the long-term benefits of soil conditioning properties of compost.
- The benefits in use of compost should be informed to farmers, who should be encouraged to partially substitute inorganic fertilizers with organic compost, as appropriate for their crop and specific soil.

The officials in the line departments such as the Agriculture & Farmers' Empowerment Department, Forest & Environment Department may be oriented for the use of "Mo Khata" and its benefits.

18. Safety Measures at Wealth Centre:

The collection, segregation, transportation, and processing activities at Wealth Centre (WC) involves exposure to contaminants and hazardous waste. To ensure healthy and safe environment at WC, the human resource engaged in activities of wealth centre are required to adopt certain safety practices at work.

18.1. Personal Protective Equipment:

The provision of Personal Protective Equipment (PPE) should be guaranteed for all sanitation workers associated in the process to ensure safety at wealth centres. While the ULB shall ensure provision of PPE, the Wealth Centre In-Charge/ Manager shall the Plant Manager shall ensure its usage by all Swachha Karmis all the time while they are handling waste. The Table-25 below shows the types of hazards and PPE to be used as a precaution.

Table 26: Type of hazards and appropriate PPEs to be used at Wealth Centre

Sl. No.	Hazards	Precaution	PPE
1.	Cuts and injuries due to presence of broken glass, sharps, needles which may lead to septic wounds and tetanus or bacterial infections by contaminated	Use of safety gloves	
2.	Exposure to fumes/ gases causing irritation of nose, throat, and lungs.	Use of particulate mask	
3.	Bacterial infection due to contaminated waste and Cuts and injuries due to presence of broken glass, sharps, needles which may lead to septic wounds and tetanus	Use of Safety shoes	
4.	Exposure of head to waste, flies and sharp objects, which may lead to infection or injury	Cap	

18.2. Fire Safety:

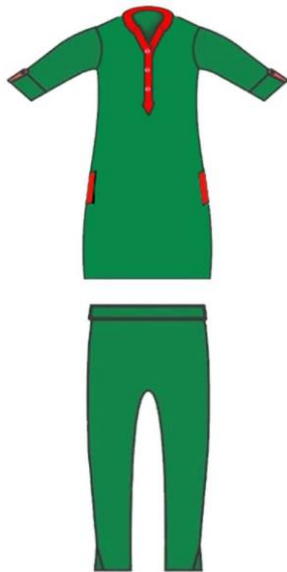
The wet waste sometimes due to methane emissions, can get fire, when there is very high temperature during summer. Also, the dry waste received at the wealth centre generally consists materials such as: wood, paper, cloth, rubber, and some plastics. The presence of these materials involves the risk of fire hazards at wealth centres. As a precaution, fire safety measures should be taken at Wealth Centres as prescribed by local fire authorities to tackle fire emergencies. The wealth centres must be equipped with a suitable fire extinguisher to be used in emergency. The human resources engaged at the Wealth Centre should be trained for such situations.

The emergency contact number for fire emergency '101' should also be known by the workers at the WC.

19. Uniform for Swachha Karmis:

The H&UD Department has prescribed standardised uniform for the community partners engaged as Swachha Karmis as mentioned in Table below –

Table-27: Specification for Uniforms

Specification for Uniforms	
Top • Short kurta • 3/4th sleeve with red piping (Roll up with bottom up) • Mandarin neck design with buttons • Two side kurta pockets (with red strip) attached to side slit • Poly Cotton Material Size – M Kurta length – 32 inch Pant • Straight cut • Bottom side slit Size – M Pant Length – 35 inch Waist round – 32 inch Hip round – 38 inch	

20. Monitoring of Wealth Centre:

20.1. State Level Monitoring Cell:

For effective supervision, monitoring, guidance, support, remedial measures, evaluation etc. and to bring up these “Wealth Centres” i.e., the MCCs and MRFs to an ideal performance level and to maintain the standard, improvise upon the experience gathered, a State Level Monitoring Cell has been constituted by this Department. (HUDD, 2019).

20.2. AMA SAHARA App:

The mobile app “AMA SAHARA” is a citizen engagement and performance management app created for Department of Housing and Urban Development, Government of Odisha.

The app facilitates

- citizens to post requests for garbage pickups and raise complaints if they are not happy with solid waste management services.

- facilitates government officials, sanitation workers and facility managers input data on the daily basis helping government to monitor the performance of the processes and solid waste management in the State of Odisha.

The app can be used at every node of the solid waste management value chain starting from the waste deposition at the wealth centre.

- The amount of dry waste and wet waste received at the wealth centre is weighed and the corresponding quantity is entered in the “AMA SAHARA” app by the Swachha Karmi duly checked by the In-charge of the wealth centre.
- The daily production of the compost and its sale is also recorded in the app.
- The quantity of recyclables sorted out from the dry waste, quantity of non-recyclables and income generated from the sale of recyclables is also entered in the app.
- Apart from this the app is also used for user fee collection from households for solid waste management service based on which monthly incentives shall be calculated for the Plant Managers.

The app helps in periodic assessment of performance and improve quality of service. The state level monitoring cell “Suo Motu” uses the app for constant monitoring and smooth functioning of the wealth centres.

Table 28: Frequency of Reporting in Ama Sahara App

Sl. No.	Type of data to be reported through Ama Sahar App	Frequency of data filling in the App
1	Wet Waste Received	Daily
2	Mo Khata Ready-to-sale	Daily
3	Mo Khata Sold	On the date of sale
4	Dry Waste Received	Daily
5	Segregated Waste	Daily
6	Recyclable Sold	On the date of sale
7	Non-recyclable Disposed	On the date of disposal
8	Sanitary waste Disposed	Daily
9	Domestic Hazardous waste disposed	On the date of disposal
10	E-waste disposed	On the date of disposal
11	Disbursal of incentive	Monthly
12	Attendance	Daily
13	Collection of user fee	Daily
14	Quantity of waste collected	Monthly
15	Waste collected from commercial units & bulk waste generators including institutions and offices	Monthly

21. Incentive Framework:

1. The Swachha Sathi/ Plant Manager shall be deployed on an incentive basis only.
2. The Sanitation Expert/ Sanitation Intern shall prepare a consolidated report on the performance deliverables upto 25th of every month and submit the report to the ULB by the last date of that month. All reports should be certified and submitted to ULB by 27th of every month/by 25th in the month of February.
3. Incentive will be released based on the performance of the Swachha Sathis/Plant Managers on or before 7th of every month.

21.1. Incentive framework for Swachha Sathi:

Table-29: Monthly incentive framework against primary performance indicator;

Sl. No.	Parameter	Eligibility	Incentive in Rs.	Means of Verification	Verification Responsibility	Replacement of Swachha Sathi
1	User Fee Collection on monthly demand from HHs/ Commercial	<50%	0	Ama Sahara App/ ULB Data	Sanitation Expert	Consistent non-performance in the 2 parameters over a period of 3 months.
		50-60%	2000	-do-	Sanitation Expert	
		61-80%	2000 + 1000	-do-	Sanitation Expert	
		> 80%	2000 + 2000	-do-	Sanitation Expert	
2	Garbage Vulnerable Points (GVP)	GVP Free	2000	Declaration by Swachha Sathi and validation of GVP status by Ward Level Committee (WLC) and submission to ULB by 27 th of each month. In case of non-receipt of report from WLC by the scheduled date, Officer in-charge of Sanitation shall verify the status for the concerned wards and release the incentive accordingly.	Officer in-charge of Sanitation	

In addition to the above indicators and incentives, Swachh Sathis along with Swachh Sabari Drivers and helpers can be paid additional incentives collectively based on the overall performance of the collection & transportation mechanism as given in Table-2.

The Swachha Sathis can also procure “Mo Khata” at a discounted rate from the wealth centre and sell either through outlets or doorstep delivery to Households, Commercial Units and Bulk Waste Generators including institutions and offices.

Table-30: Additional monthly incentive framework against secondary performance indicator;

Sl. No.	Parameter	Eligibility	Incentive in Rs.	Means of Verification	Beneficiary	Verification Responsibility
1	Quantity of Waste Collection (1 KG x No. of assigned HH/Com.x 25 Days)	> 80%	Rs.3000/- (@Rs.100 0/- each)	1. Daily Trip Register 2. Ama Sahara App	1. Swachha Sathi 2. Swachha Karmi-only Swachha Sabari Driver) 3. Swachha Karmi-(Swachha Sabari Helper only)	Sanitation Expert
2	Segregated Waste received in a month at the Wealth Centre	> 80%	Rs.3000/- (@Rs.100 0/- each)	Ama Sahara App or Daily Incoming Register	1. Swachha Sathi 2. Swachha Karmi (-Swachha Sabari Driver only) 3. Swachha Karmi (-Swachha Sabari Helper only)	Sanitation Expert
3	Door-step Sale of Mo Khata to Households, Commercial Establishments and Bulk Waste Generators including institutions and offices.		"Mo Khata" at 20% discounted rate	1. "Mo Khata" stock register 2. Ama Sahara App	1. Swachha Sathi 2. Swachha Karmi (-Swachha Sabari Driver only) 3. Swachha Karmi-(Swachha Sabari Helper only)	

21.2. Incentive framework for Plant Manager:

Table 31: Monthly performance incentive framework;

Sl. No.	Indicator	Eligibility	Incentive in Rs.	Means of Verification	Verification responsibility	Replacement
1.	Collection & Transportation:					Consistent non-performance in over a period of 3 months.
1.1	Waste collection from Commercial Units & Bulk Waste Generators including institutions & offices	> 80% of the estimated collection (As per SoP)	3000	Ama Sahara Register	Sanitation Intern	
1.2	Fleet management & monitoring of BoV/LCV (monitoring the movement of vehicles for collection of waste and maintain the register)	Proper maintenance of the register	1000	Vehicle Register	Sanitation Intern	

2.	Functioning of MCC/MRF:				
2.1	MCC: Regular maintenance and cleaning of drainage channel, periodic desludging of leachate tank/pit	Minimum 4 times in a month/when it is full	2000	Maintenance register	Sanitation Intern
	MRF: 100% sorting of dry waste received same day	100% sorting on a 7 day (weekly basis)		Sorting register	Sanitation Intern
2.2	Safety and Hygienic Condition of MCC/MRF (Cleanliness of MCC/MRF, functional separate toilets, use of PPE)	30 days cleanliness (Rs.100/- deduction for each day of non-compliance)	2000	Daily cleaning chart	Sanitation Intern
2.3	HR management of Swachha Karmis at MCC/MRF and ensuring each shift of 8 hour/day of Wealth Centre	Maintenance of HR Register and Rs 100 deduction for each day of non-compliance	2000	Ama Sahara/HR Register	Sanitation Expert
3.	Sale of Mo Khata/Recyclable & non-recyclables:				
3.1	Timely packaging & lifting/sale of "Mo Khata" for MCC /recyclables & non-recyclables for MRF	Regular clearance of materials from the MCC/MRF	2000	Ama Sahara / Register	Sanitation Expert
	Total (1 +2 +3)		12,000		

21.3. Incentive framework for Swachha Karmi:

Swachha Karmis are entitled to monthly incentive as may be notified by Department from time to time.

The ULBs will ensure that in addition to the monthly incentives, the Swachha Karmi engaged from SHGs avail their basic entitlements listed in the Table below.

Table 32: Entitlements of Swachh Karmis

Sl. No.	Entitlements	
1	Weekly one-day paid leave	The Plant Manager must ensure that the daily waste management activities are not affected due to absence of multiple Swachh Karmis on a particular day. Hence, all leaves are to be granted after careful assessment.
2	Group Personal Accidental Insurance	The H & UD Department has covered all the Swachh Karmis of MCC and MRF under the Group Personal Accident Insurance of Rs. 2 lakh per person w.e.f. 2023, which is subjected to renewal every year.
3	Health check-up	Quarterly by Urban PHC
4	Personal Protective Equipment (discussed with details under the "Safety Measures" in this document)	To be provided by ULB for use by the Swachh Karmis all the time while handling waste
5	Lounge with individual locker facility and First Aid box	Common Lounge to be provided at each wealth centre for use by Swachh Karmis for dining and storing belongings
6	Four sets of uniform, gloves head scarf etc.	

22. Modalities for Payment:

- 22.1. ULBs to issue a one-time Standing Instruction to the Banks concerned for remittance of the fixed monthly incentive to the Aadhaar linked Savings Bank Account of the Swachh Karmi concerned associated with solid waste processing in Wealth Centres on 5th of each month, unless a contrary advice to "Stop payment" is issued by the ULB before that date, whenever situation warrants.
- 22.2. Performance Incentive as stipulated in preceding paragraphs may be remitted to the Aadhaar linked Savings Bank Account of the Swachh Karmi concerned on quarterly basis.
- 22.3. A standard format is developed and attached for issuing to the Bank Authority as STANDING INSTRUCTIONS is annexed herewith at ANNEXURE-B (along with Format "SCHEDULE OF MONTHLY INCENTIVE" as Annexure-B-1)
- 22.4. If it is found that, amount payable is to be withheld for lack of delivery of services or absent or for any other valid reason(s) on administrative grounds; an instruction to that effect may be issued to the Bank concerned before 5th and subsequently upon rectification / compliance, the amount payable may be released.

- 22.5. Any amount that needs to be deducted or recovered from the payable or already paid may be affected in subsequent months also. Hence for want of verification of absentee statement or on account of services delivered no undue delay should happen in monthly payment to Swachh Karmis.
- 22.6. The concerned E.O. and Accountant will be held personally liable for any delay in releasing payment to the Community Partners.
- 22.7. It is the responsibility of the E.O to ensure necessary coordination with the relevant Banks so as to achieve the objective of timely payment to Community Partners.

23. Collection of User Fee:

- 23.1. ULB shall authorize Swachha Sathis to collect User fee and arrange wide publicity on the initiative in print and electronic media and other means.
- 23.2. Each Swachha Sathi shall collect user fee through digital mode only from the Households/Commercial Units/Bulk Waste Generators including institutions and offices through a QR Code linked to ULBs bank account.
- 23.3. ULB shall register the Swachha Sathis in AMA Sahara App. Sanitation Expert as Maker shall register the Swachha Sathis in the application and Officer In-Charge of Sanitation as Checker shall approve the same. Unique ID for each Swachha Sathi shall be generated after the successful registration of the Swachha Sathis as per the following coding formula;

Table 33: Coding of Swachha Sathis:

1 st 2 Digit	Programme Code	Swachha Sathi	SS
2 nd 6 Digit	ULB Census Code	ULB Census Code	xxxxxx
3 rd 3 Digit	SHG Code	Sl. No.	001
4 th 3 Digit	Swachha Sathi Code	Sl. No.	001

- 23.4. Hands-on training shall be given to the Swachha Sathis followed by On-site Training.

23.4.1. Gap funding

The Wealth Centres are designed to be self-sustainable waste management units. However, till the system is stabilised and the earning from the Wealth Centre (including the sanitation user fees) fully meets the expenditures incurred on operation and maintenance of the Wealth Centres, the ULB may remit the incentives to the community partners as agreed upon from its own fund, if required.

24. Monitoring and Evaluation:

1. Sanitation Expert/Sanitation Intern shall monitor the performance of the Swachha Sathis/Plant Managers respectively every month
2. ULB level Swachhata Committee shall evaluate the performance of Swachha Sathi/ Plant Manager on quarterly basis
3. “Ama Sahara”/Database developed shall be utilized for daily, weekly, monthly and quarterly performance of Swachha Sathis/Plant Managers by ULBs. The broad areas of monitoring shall include the following:
 - a. Activity and meeting of targets of Swachha Sathis/Plant Managers
 - b. Coverage of all Households/Commercial Units/Bulk Waste Generators including institutions and offices for user fee collection at ward level
 - c. Collection efficiency of segregated waste at ward level/Wealth Centre level
4. State shall monitor the progress and achievement for performance evaluation of Sanitation Experts/Sanitation Interns based on inputs from ULB team.

Table 34: List of registers:

Registers	Responsible Person	Annexure
HR register	Plant Manager	Annexure 1
Cleanliness & Hygiene Chart	Plant Manager	Annexure 2
Sorting register	Plant Manager	Annexure 3
Incident register	Plant Manager	Annexure 4
Fleet Management register	Plant Manager	Annexure 5
Daily Trip register	Plant Manager	Annexure 6
Recyclable/Non-Recyclable Waste Sale Register	Plant Manager	Annexure 7
Mo Khata register	Plant Manager	Annexure 8
Maintenance register	Swachha Sabari Driver	Annexure 9
User Fee register	Plant Manager	Annexure 10

Annexure 1

HR Register						
Sl. No	Date	Name of Swachh Karmi	Present/Absent	In-time	Out-time	Leave

Annexure 2

Cleanliness & Hygiene Chart								
Sl. No.	Activities	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
1	Secondary segregation area cleaned							
2	Vehicle Parking Area and Charging Points cleaned							
3	Vehicle Cleaning Station cleaned							
4	Leachate network cleaned							
5	Drainage line cleaned							
6	Change Room/Lounge cleaned							
7	Toilet cleaned							
8	Swachh Karmi wearing PPE Kits							
9	Swachh Sabari Driver and Helper wearing PPE Kits							

Annexure 3

Sorting Register				
Name of the Plant Manager:			MCC/MRF ID:	
Date	Quantity of Incoming Waste	Quantity of Source Segregated Waste (MT)	Quantity of Unsegregated Waste (MT) to be sorted	Quantity of Waste (MT) to be sorted

Annexure 4

Incident Register				
Name of Plant Manager:				
MCC/MRF ID:				
Date of Incident	Type of Incident	Whether reported to Nodal Officer	Action Taken	Current Status

Annexure 5

[illegible]

Annexure 6

[illegible]

Annexure 7

[illegible]

Annexure 8

[illegible]

Annexure 9

Maintenance Register				
Date:			Name of the Plant Manager:	
Indicators				
Sl. No	Machineries	Functional (Y/N)	Action taken	Remarks
1	Digital weighing bridge			
2	Shredding Machine			
3	Conveyor Belt			
4	Sieving machine			
5	Fly Traps			
6	Baling Machine			
7	Turbo Exhaust Fans installed			
8	Jet Speed Cleaning Machine			
9	All BOVs			
10	All LCVs			
11	Water Purifier			
12	Napkin incinerator			
13	Sanitary Pads vending machine			
	Cleanliness (Periodic)			
14	Leachate network			
15	Drainage line cleaned			
	Fittings and fixtures are functional/dysfunctional			
16	Electric Supply (Yes/No)			
17	Water Supply(Yes/No)			
18	Fans in the compost drying yard and office room functional (Yes/No)			
19	Any faults in electrical switch boards?			
20	Office accessories like Table, Chair, Almirah			
21	Compost bag stitching machine			

Annexure 10

Swachh Sathi & Household Category Wise User Fee Collection by <<ULB Name>>

From Date:		To Date:						
ULB Name	Swachh Sathi Name	Swachh Sathi ID	Ward Number	Household Category	Total Households	User Fee Target (in Rs./-)	Households Covered with User Fee	User Fee Collected (in Rs./-)

Monthly Payout Summary of Swachh Sathis

ULB Name:				
Payout for the Month of:				
Name of Swachh Sathis	User Fee Target	User Fee Collected	% of Collection	Incentive Amount

Monthly Payout of Swachh Sathi

Payout for the Month of				
ULB Name:				
Name of the Swachh Sathi:				
Swachh Sathi ID:				
SHG Name:				
SHG ID:				
Household Category Wise User Fee Collection				
Household Category	No. of Unit	User Fee Target	Covered with User Fee	User Fee Collected
Residential				
Commercial				

Industry (MSME)				
Kalyan Mandap				
Hostel				
Temple				
School/College				
Dharmasala				
Other Categories.....				
Other Categories.....				
Other Categories.....				
Total:				

Incentive to be Released based on Total Amount Collected:	(____%)	
--	---------	--

Annexure 11

Declaration Form on GVP Status

ULB Name:		Reporting Month:	
Name of the Swachha Sathi:		Date of Submission of the report:	
Ward No.	Location of the GVP(s)	Status of the GVP (Cleared/Present)	
Submitted by		Verified by	
Name of the Swachha Sathi:		Name of the Councillor/Ward Officer:	
Signature of the Swachha Sathi:		Signature of the Councillor/Ward Officer:	

Annexure 12

1st Notice

This is to bring to your notice that during our monthly evaluation of performance of Swachha Sathis/Plant Manager, MCC/ Plant Manager, MRF, the <<Name>> working as Swachha Sathis/Plant Manager, MCC/ Plant Manager, MRF in MCC/MRF <<ID>> has been found not to be performing as per the performance indicators.

The details of non-performance are as follows:

1. _____
2. _____

<<Name>> will be imparted training and provided handholding support for one month for improvement in performance. You are therefore requested to intimate <<Name>> about the notice and ensure that there is improvement in non-performance.

Nodal Officer
Wealth Centre
ULB Name

Copy to Addl. Commissioner/EO, _____ Corporation/Municipality/NAC for favour of information and necessary action.

Nodal Officer
Wealth Centre
ULB Name

To

(Name & address of the SHG/TG Group/WPG)

Annexure 13

2nd Notice

This is in continuation to our previous Notice vide letter nodated xxxxxx concerning the non-performance of <<Name>> working as Swachha Sathis/Plant Manager, MCC/ Plant Manager, MRF in MCC/MRF <<ID>>. Despite our extension of handholding support the <<Name>> working as Swachha Sathis/Plant Manager, MCC/ Plant Manager, MRF in MCC/MRF <<ID>> continueto be non-performing during the month of XXXXXX.

This is a show cause for the continuous performance and also to provide one more months' time to improve performance failing which we will issue 7-day notice for replacement of the Swachha Sathi/ Plant Manager.

You are therefore requested to intimate <<Name>> and submit a reply of the show cause notice and ensure that there is improvement in non-performance.

Nodal Officer

Wealth Centre

ULB Name

Copy to Addl. Commissioner/EO, _____Corporation/Municipality/NAC for favour of information and necessary action.

Nodal Officer

Wealth Centre

ULB Name

To

(Name & address of the SHG/TG Group/WPG)

Annexure 14

3rd Notice

This is in continuation to our previous Notice vide letter nodated xxxxxx and second notice/Show cause vide No._____dated_____concerning the non-performance of <<Name>> working as Swachha Sathis/Plant Managers in MCC/MRF <<ID>>. Despite extending handholding support, <<Name>> working as Swachha Sathis/Plant Manager in MCC/MRF <<ID>> continue to be non-performing continuously for the months of_,_&_.

This is the final notice for replacement of the Swachha Sathi/ Plant Manager and provide a suitable replacement meeting the eligibility criteria within a period of seven working days to the ULB communicating the details of the new Swachha Sathi/ Plant Manager and the expected date of joining. Failure to provide suitable replacement within 15 days will make the Agreement with the SHGs/TGs/WPGs liable for terminate and ULB will take steps for deployment of Swachha Sathi/ Plant Manager from any other suitable SHGs/TGs/WPGs.

Nodal Officer

Wealth Centre

ULB Name

Copy to Addl. Commissioner/EO, _____Corporation/Municipality/NAC for favour of information and necessary action.

Nodal Officer

Wealth Centre

ULB Name

To

(Name & address of the SHG/TG Group/WPG)

Annexure 15

FORMAT FOR STANDING INSTRUCTIONS (SI) TO BE ISSUED TO BANKS

Office of the Corporation / Council

(Name of the ULB)

From

Commissioner/ Additional Commissioner/ Joint Commissioner/

Deputy Commissioner/ DDO/ Executive Officer ----- (Name of the ULB)

To

The Bank Manager,

----- (Name of the Bank)

----- (Name of the Branch)

----- (Address)

Sub: Issue of Standing Instruction for release of Honorarium of Swachh Karmis associated in processing of Solid waste in the Wealth Centres.

Madam/ Sir,

I am to intimate that monthly incentives of Swachh Karmis associated with Operation and Management of MCCs and MRFs established within the jurisdiction of this Urban Local Body are being paid through your Bank.

I authorise you to debit a sum of Rs. _____ (Amount in Rupees in words) on **5th of each month** from the Bank Account No. _____ of this ULB maintained in your Bank as per the list attached herewith as ANNEXURE-B-1, unless "STOP PAYMENT" instruction in writing by the undersigned or the authorised officer is issued in this regard.

Yours faithfully,

Commissioner/Additional Commissioner/ Joint
Commissioner/ Deputy Commissioner/ DDO/ Executive Officer
----- (Name of the ULB)

Annexure 16

“SCHEDULE OF MONTHLY INCENTIVE”

Sl. No.	Name of the Swachh Karmi	Name and Code of the Wealth Centre	Amount of monthly Honorarium including R&H allowance	Name of the Bank of the Beneficiary	Bank Account Number of the Beneficiary	IFS Code of the Bank
1	2	3	4	5	6	7

N:B: Changes in the aforesaid List required, if any, will be communicated from time to time.

Commissioner/ Additional Commissioner/ Joint
 Commissioner/ Deputy Commissioner/ DDO/ Executive Officer
 ----- (Name of the ULB)

Annexure 17

Agreement for O&M of MCC

This agreement, is made on this day, the _____ day of the month of _____, year, between <Name of ULB>, <District Name>, Odisha represented by the Commissioner or Executive Officer (As the case may be), here-in-after referred to as “1st Party”, which expression shall include its respective successors and permitted assignees of the part.

AND

<Name of the SHG/Transgender Group/Waste Picker Group>, <Ward No>, <ULB>, <District>, Odisha represented by the President/Secretary, here-in-after called the “2nd Party” which expression shall include its respective successors and permitted assignees.

WHEREAS, <ULB Name> is the Nodal Agency to implement Solid Waste Management in <ULB Name>. The biodegradable waste is transported from the households, commercial units, institutions and offices to the Micro Composting Centre (MCC) where it is processed to generate organic compost called “Mo Khata”. The state has decided to improve the SWM value chain in partnership with the urban SHG/Transgender Group/Waste Picker Group (SHG/TG Group/WPGs), which will in turn create livelihood opportunity for their members.

WHEREAS, the 1st Party intends to sign an agreement with 2nd Party for the Operation and Maintenance of MCC by deploying Swachha Karmi (s) at <Name of MCC>, <Ward No__> as per the following terms and conditions;

I. DURATION OF THE AGREEMENT

1. The agreement shall be in force for two (2) years from the date of signing of the agreement unless otherwise decided by ULB.
2. The agreement, at the discretion of ULB, is likely to be extended for another one (1) year subject to satisfactory performance by the SHG/TG Group/WPG.

II. ROLES AND RESPONSIBILITIES OF THE ULB

1. ULB shall ensure the MCC is in functional and maintained condition with overall standard amenities, safety measures, access to electricity and water connection, required equipment or materials and necessary infrastructure.
2. ULB shall issue identity card and uniforms to the members of the SHG/TG GROUP/WPG deployed at the MCC.
3. ULB shall provide and facilitate trainings for SHG/TG GROUP/WPG members on the Standard Operating Procedures (SOPs) for operation and maintenance of the MCC. The trainings shall not be limited to use of equipment and machinery, but shall include step by step procedure on biodegradable waste management at MCC, personal health and safety, record keeping, etc.
4. ULB will appoint a Nodal officer, who shall be responsible for smooth functioning/ monitoring of the facility and creating a coordination system.
5. ULB holds the responsibility to collect and transport segregated dry waste to the MCC.
6. ULB shall coordinate and supervise the works of the SHG/TG GROUP/WPG on regular basis. ULB shall provide day to day support to the SHG/TG Group/WPG for any technical issues they may face.
7. ULB shall convene meetings periodically with all the stakeholder to discuss the challenges and issues faced by the SHG/TG GROUP/WPG and others during operation and management of MCC and address those following due process.
8. ULB shall be responsible for selling of Mo Khata. The quantity of Mo Khata sold and the amount received are to be recorded in register and reported through 'Ama Sahara App' on real time.
9. The ULB shall pay the SHG/TG Group/WPG for buying consumable for Effective Microorganism (EM).
10. The ULB shall provide Personal Protective Equipment (PPE) and ensure availability of "First Aid Kit" at the MCC.
11. ULB shall cover the Swachh Karmis deployed by the SHG/TG Group/WPG under group personal accident insurance.
12. ULB shall conduct regular health checkup of the SHG/TG GROUP/WPG members working at the MCC.

III. ROLES AND RESPONSIBILITIES OF THE SHG/TG GROUP/WPG

1. The SHG/TG Group/WPG shall submit the list of the eligible member(s) to be deployed as Swachha Karmis.
2. SHG/TG Group/WPG shall always ensure that the member(s) deployed as Swachha Karmis are trained by ULB before commencement of work.
3. The SHG/TG Group/WPG shall provide suitable replacement of Swachha Karmi (s) if any of the Swachha Karmis discontinue or found to be not performing as per prescribed roles and responsibilities.

IV. ROLES AND RESPONSIBILITIES OF THE SWACHHA KARMIS

1. Swachha Karmis shall be responsible for operations and maintenance of the MCC as per the SOP.
2. Swachha Karmis shall be responsible for reporting about any defects, deficiencies and malfunctioning in the machinery at the MCC.
3. Swachha Karmis shall keep the MCC clean and tidy.

V. GENERAL TERMS AND CONDITIONS

1. Binding Effect:

This Agreement shall be binding on and, to the extent such successors or assigns are permitted, inure to the benefit of the successors, and assigns of the ULB and the SHG/TG Group/WPG.

2. ULB's Right to Inspect:

The ULB or its designated / authorized representative / employees shall have the right to inspect the SWM status and records on user fee collection at any time or any portion thereof.

3. Amendment:

This Agreement may be amended only in writing duly agreed upon by the ULB and the SHG/TG Group/WPG.

4. Notices:

Any notice or other communication to be given by any Party to the other Party under or in connection with the matters contemplated by this Agreement shall be in writing and shall be delivered to persons as follows:

4.1. For the ULB:

Name:

Address:

Telephone:

E-mail:

4.2. For the SHG/TG Group/WPG:

Name:

Address:

Telephone:

E-mail:

5. Miscellaneous:

5.1. The terms of this Agreement shall not be construed to constitute a

partnership, joint venture, or employer/employee relationship between the Parties.

- 5.2. This Agreement cannot be further sub-contracted; however, the SHG/TG Group/WPG can allow any member of the group to participate in training programme without any claim.
- 5.3. In the event of any provision of this Agreement being held or becoming invalid, unenforceable, or illegal for any reason, by both the Parties and a proceeding drawn to that effect and duly signed, this Agreement shall remain otherwise in full force apart from the said provision which shall be deemed to be deleted / modified, as the case may be. The Parties shall however attempt to replace the deleted provision with a legally valid provision that reflects the same purpose of the deleted provision to the greatest extent possible.
- 5.4. For non-adherence to any instructions / conditions / obligations arising out of the agreement, the ULB may issue written intimation to the SHG/TG Group/WPG clearly stating therein the violations made / aspects of non-observance, etc. and to rectify the same within a specified period to be indicated in the said written intimation. The SHG/TG Group/WPG should comply with the same and report to the ULB.

6. Applicable Law:

Applicable Law means the laws and any other instruments having the force of law in Odisha as they may be issued and in forced, from time to time.

7. Arbitration:

- 7.1. The Parties agree that the avoidance or early resolution of disputes is crucial for a smooth execution of the Agreement and the success of the assignment. The Parties shall use their best efforts to settle amicably all disputes arising out of or in connection with this Agreement or its interpretation.
- 7.2. Municipal Commissioner/ Project Director, District Urban Development Agency of the <district> shall act as the Mediator/Arbitrator for resolving any issue between the ULB and the SHG/TG Group/WPG within 30 days from date of receipt of the written complaint.
- 7.3. In case of failure in amicable settlement, the dispute shall be referred to the Collector of the district for redressal.
- 7.4. In case the dispute subsists, Parties may avail the forum under the Arbitration and Conciliation Act, 1996 read with Amendment Act, 2015 and shall be held in <Name of the town>, <Name of the district> India.
- 7.5. The Agreement and the rights and obligations of the Parties shall remain in full force and effect, pending the Award in the arbitration proceedings.

8. Force majeure:

Earthquakes, floods, fires, explosions, acts of God, acts of State, war, terrorism,

action of any Government including changing governmental policy or any other situation, which is beyond the reasonable control of a Party may be treated as the Force Majeure event. For avoidance of doubt, it is clarified that, willful and intentional omissions, or commissions in relation to any Party shall not be considered as a Force Majeure event.

Neither of the Parties shall be entitled to claim benefit of a Force Majeure event unless it has intimated the other Party of occurrence of such event within 48 hours from the occurrence thereof, indicating there in the steps it is taking or intending to take to mitigate the effect of such Force Majeure event on performance of its obligations under this Agreement.

In the event of Force Majeure, the respective obligations and responsibilities shall remain suspended. If the Force Majeure continues for a period of more than 30 days, the Parties can re-negotiate this Agreement in good faith and if the Parties do not reach any consensus within 15 days from the end of such 30 days period referred to above, this Agreement shall automatically stand terminated on such 45th day.

9. Recovery of cost for damage / loss of articles:

The ULB may recover the cost of items [equivalent to the market value of lost item(s)] or other assets if any in case of loss.

10. Termination of Agreement:

The Agreement may be terminated by either party under the following situation:

- 10.1. In case the SHGs/TGs/WPGs is found to be violating the terms of the agreement or found to be not adhering to the terms during inspection by ULB, the ULB shall have the right to terminate the Agreement by giving a written notice of not less than 30 days. During this period, the SHGs/TGs/WPGs shall be provided due opportunity to take corrective actions within 15 days from the date of notice with further extension, if deemed required not exceeding 7 days.
- 10.2. In case SHGs/TGs/WPGs expresses unwillingness/unable to continue with the Agreement, then the SHGs/TGs/WPGs shall have the right to get withdrawn from the Agreement on giving a 30 days' notice to ULB.
- 10.3. In case of dissolution of SHGs/TGs/WPGs, Agreement shall stand terminated on receipt of copy of the resolution stating the dissolution of SHGs/TGs/WPGs.
- 10.4. The termination of the SHGs/TGs/WPGs shall be done by the ULB with due approval from the ULB level Swachhata Committee.

10. Indemnity:

To the fullest extent permitted by applicable law and professional regulations, the SHG/TG Group/WPG shall indemnify the ULB against all claims by third parties and resulting liabilities, losses, damages, costs, and expenses arising out of third party's use of the facilities.

11. Interpretation:

- 11.1. All words and expressions used in this Agreement shall, where relevant, have the meaning respectively assigned to it under any Act or any other law for the time being in force, and in the absence thereof, the Parties may suitably discuss and adhere to the modalities prescribed thereafter.
- 11.2. All words in singular shall be deemed to connote their respective plurals and vice-versa unless the context suggests otherwise.
- 11.3. The words "include" and "including" are to be construed without limitation.
- 11.4. The Appendixes to this Agreement form an integral part of this Agreement and shall be interpreted accordingly. However, the Appendix while providing information both relevant and with respect to certain aspects of the Agreement, shall be separate from it, and not read as integral to the Agreement.

12. Appendixes:

- 1) Standard Operating Procedure, SOP (Appendix-3)

Annexure 18

Agreement for O&M of MRF

This agreement, is made on this day, the _____ day of the month of _____, year, between <Name of ULB>, <District Name>, Odisha represented by the Commissioner or Executive Officer (As the case may be), here-in-after referred to as “1st Party”, which expression shall include its respective successors and permitted assignees of the part.

AND

<Name of the SHG/Transgender Group/Waste Picker Group>, <Ward No>, <ULB>, <District>, Odisha represented by the President/Secretary, here-in-after called the “2nd Party” which expression shall include its respective successors and permitted assignees.

WHEREAS, <ULB Name> is the Nodal Agency for Municipal Solid Waste Management (SWM) in <ULB Name>. The non-biodegradable wastes are transported from the households, commercial units, institutions and offices to the Material Recovery Facility (MRF) for sorting and safe disposal. The sorted recyclable wastes are sold to the empaneled recyclers/ agencies and the non-recyclables and inert wastes are safely disposed. The state has decided to improve the SWM value chain in partnership with the urban SHG/Transgender Group/Waste Picker Group (SHG/TG Group/WPGs), which will in turn create livelihood opportunity for their members.

WHEREAS, the 1st Party intends to sign an agreement with 2nd Party for the Operation and Maintenance of MRF by deploying Swachha Karmi (s) at <Name of MRF>, <Ward No__> as per the following terms and conditions:

I. DURATION OF THE AGREEMENT

1. The agreement shall be in force for two (2) years from the date of signing of the agreement unless otherwise decided by ULB.
2. The agreement, at the discretion of ULB, is likely to be extended for another one (1) year subject to satisfactory performance by the SHG/TG Group/WPG.

II. ROLES AND RESPONSIBILITIES OF THE ULB

1. ULB shall ensure the MRF is in functional and maintained condition with overall standard amenities, safety measures, access to electricity and water connection, required equipment or materials and necessary infrastructure.
2. ULB shall issue identity card and uniforms to the members of the SHG/TG GROUP/WPG deployed at the MRF.
3. ULB shall provide and facilitate trainings for SHG/TG GROUP/ WPG members on the Standard Operating Procedures (SOPs) for operation and maintenance of the MRF. The trainings shall not be limited to use of equipment and machinery, but shall include step by step procedure on dry waste management at Material Recovery Facility, personal health and safety, record keeping, etc.
4. ULB will appoint a Nodal officer, who shall be responsible for smooth functioning/ monitoring of the facility and creating a coordination system.
5. ULB holds the responsibility to collect and transport segregated dry waste to the MRF.
6. ULB shall coordinate and supervise the works of the SHG/TG GROUP/WPG on regular basis. ULB shall provide day to day support to the SHG/TG Group/WPG for any technical issues they may face.
7. ULB shall convene meetings periodically with all the stakeholder to discuss the challenges and issues faced by the SHG/TG GROUP/WPG and others during operation and management of MRF and address those following due process.
8. ULB shall be responsible for selling of the recyclables waste to the bulk waste recycler or any other party empaneled by the ULB. The quantity of recyclable sold and the amount received are to be recorded in register and reported through 'Ama Sahara App' on real time.
9. The ULB shall provide Personal Protective Equipment (PPE) and ensure availability of "First Aid Kit" at the MRF.
10. ULB shall cover the Swachh Karmis deployed by the SHG/TG Group/WPG under group personal accident insurance.
11. ULB shall conduct regular health checkup of the SHG/TG GROUP/WPG members working at the MRF.

III. ROLES AND RESPONSIBILITIES OF THE SHG/TG GROUP/WPG

1. The SHG/TG Group/WPG shall submit the list of the eligible member(s) to be deployed as Swachha Karmis.
2. SHG/TG Group/WPG shall always ensure that the member(s) deployed as Swachha Karmis are trained by ULB before commencement of work.
3. The SHG/TG Group/WPG shall provide suitable replacement of Swachha Karmi (s) if any of the Swachha Karmis discontinue or found to be not performing as per prescribed roles and responsibilities.

IV. ROLES AND RESPONSIBILITIES OF THE SWACHHA KARMIS

1. Swachha Karmis shall be responsible for operations and maintenance of the MRF as per the SOP.

2. Swachha Karmis shall be responsible for reporting about any defects, deficiencies and malfunctioning in the machinery at the MRF.
3. Swachha Karmis shall keep the MRF clean and tidy.

V. GENERAL TERMS AND CONDITIONS

1. Binding Effect:

This Agreement shall be binding on and, to the extent such successors or assigns are permitted, inure to the benefit of the successors, and assigns of the ULB and the SHG/TG Group/WPG.

2. ULB's Right to Inspect:

The ULB or its designated / authorized representative / employees shall have the right to inspect the SWM status and records on user fee collection at any time or any portion thereof.

3. Amendment:

This Agreement may be amended only in writing duly agreed upon by the ULB and the SHG/TG Group/WPG.

4. Notices:

Any notice or other communication to be given by any Party to the other Party under or in connection with the matters contemplated by this Agreement shall be in writing and shall be delivered to persons as follows:

A. For the ULB:

Name:

Address:

Telephone:

E-mail:

B. For the SHG/TG Group/WPG:

Name:

Address:

Telephone:

E-mail:

12. Miscellaneous:

1. The terms of this Agreement shall not be construed to constitute a partnership, joint venture, or employer/employee relationship between the Parties.
2. This Agreement cannot be further sub-contracted; however, the

SHG/TG Group/WPG can allow any member of the group to participate in training programme without any claim.

3. In the event of any provision of this Agreement being held or becoming invalid, unenforceable, or illegal for any reason, by both the Parties and a proceeding drawn to that effect and duly signed, this Agreement shall remain otherwise in full force apart from the said provision which shall be deemed to be deleted / modified, as the case may be. The Parties shall however attempt to replace the deleted provision with a legally valid provision that reflects the same purpose of the deleted provision to the greatest extent possible.
4. For non-adherence to any instructions / conditions / obligations arising out of the agreement, the ULB may issue written intimation to the SHG/TG Group/WPG clearly stating therein the violations made / aspects of non-observance, etc. and to rectify the same within a specified period to be indicated in the said written intimation. The SHG/TG Group/WPG should comply with the same and report to the ULB.

13. Applicable Law:

Applicable Law means the laws and any other instruments having the force of law in Odisha as they may be issued and in forced, from time to time.

14. Arbitration:

1. The Parties agree that the avoidance or early resolution of disputes is crucial for a smooth execution of the Agreement and the success of the assignment. The Parties shall use their best efforts to settle amicably all disputes arising out of or in connection with this Agreement or its interpretation.
2. Municipal Commissioner/ Project Director, District Urban Development Agency of the <district> shall act as the Mediator/Arbitrator for resolving any issue between the ULB and the SHG/TG Group/WPG within 30 days from date of receipt of the written complaint.
3. In case of failure in amicable settlement, the dispute shall be referred to the Collector of the district for redressal.
4. In case the dispute subsists, Parties may avail the forum under the Arbitration and Conciliation Act, 1996 read with Amendment Act, 2015 and shall be held in <Name of the town>, <Name of the district> India.
5. The Agreement and the rights and obligations of the Parties shall remain in full force and effect, pending the Award in the arbitration proceedings.

15. Force majeure:

Earthquakes, floods, fires, explosions, acts of God, acts of State, war, terrorism,

action of any Government including changing governmental policy or any other situation, which is beyond the reasonable control of a Party may be treated as the Force Majeure event. For avoidance of doubt, it is clarified that, willful and intentional omissions, or commissions in relation to any Party shall not be considered as a Force Majeure event.

Neither of the Parties shall be entitled to claim benefit of a Force Majeure event unless it has intimated the other Party of occurrence of such event within 48 hours from the occurrence thereof, indicating there in the steps it is taking or intending to take to mitigate the effect of such Force Majeure event on performance of its obligations under this Agreement.

In the event of Force Majeure, the respective obligations and responsibilities shall remain suspended. If the Force Majeure continues for a period of more than 30 days, the Parties can re-negotiate this Agreement in good faith and if the Parties do not reach any consensus within the 15 days from the end of such 30 days period referred to above, this Agreement shall automatically stand terminated on such 45th day.

16. Recovery of cost for damage / loss of articles:

The ULB may recover the cost of items [equivalent to the market value of lost item(s)] or other assets if any in case of loss.

17. Termination of Agreement:

The Agreement may be terminated by either party under the following situation:

1. In case the SHGs/TGs/WPGs is found to be violating the terms of the agreement or found not to be adhering to the terms during inspection by ULB, ULB shall have the right to terminate the Agreement by giving a written notice of not less than 30 days. During this period, the SHGs/TGs/WPGs shall be provided due opportunity to take corrective actions within 15 days from the date of notice with further extension, if deemed/required not exceeding 7 days.
2. In case SHGs/TGs/WPGs expresses unwillingness/unable to continue with the Agreement, then the SHGs/TGs/WPGs shall have the right to get withdrawn from the Agreement on giving a 30 days' notice to ULB.
3. In case of dissolution of SHGs/TGs/WPGs, Agreement shall stand terminated on receipt of copy of the resolution stating the dissolution of SHGs/TGs/WPGs.
4. The termination of the SHGs/TGs/WPGs shall be done by the ULB with due approval from the ULB level Swachhata Committee.

18. Indemnity:

To the fullest extent permitted by applicable law and professional regulations, the SHG/TG Group/WPG shall indemnify the ULB against all claims by third parties and resulting liabilities, losses, damages, costs, and expenses arising out of third party's use of the facilities.

19. Interpretation:

1. All words and expressions used in this Agreement shall, where relevant, have the meaning respectively assigned to it under any Act or any other law for the time being in force, and in the absence thereof, the Parties may suitably discuss and adhere to the modalities prescribed thereafter.
2. All words in singular shall be deemed to connote their respective plurals and vice-versa unless the context suggests otherwise.
3. The words "include" and "including" are to be construed without limitation.
4. The Appendixes to this Agreement form an integral part of this Agreement and shall be interpreted accordingly. However, the Appendix while providing information both relevant and with respect to certain aspects of the Agreement, shall be separate from it, and not read as integral to the Agreement.

Appendixes:

- 2) Standard Operating Procedure, SOP (Appendix-3)

Annexure 19

AGREEMENT FOR DEPLOYMENT OF SWACHHA SATHI (S)

This agreement, is made on this day, the _____ day of the month of _____, year, between <Name of ULB>, <District Name>, Odisha represented by the Commissioner or Executive Officer (As the case may be), here-in-after referred to as “1st Party”, which expression shall include its respective successors and permitted assignees of the part.

AND

<Name of the SHG/Transgender Group/Waste Picker Group>, <Ward No>, <ULB>, <District>, Odisha represented by the President/Secretary, here-in-after called the “2nd Party” which expression shall include its respective successors and permitted assignees.

WHEREAS, <ULB Name> is the Nodal Agency to implement Solid Waste Management in <ULB Name>. It is dedicated to improve collection and transportation of municipal waste in the city by involving Community Based Organizations (CBOs) to create livelihood opportunity for the urban SHG/TG Group/WPGs.

WHEREAS, the 1st Party intends to sign an agreement with 2nd Party to support collection and transportation of Municipal Solid Waste by deploying Swachha Sathi (s) for <Ward No_> as per the following terms and conditions;

I. DURATION OF THE AGREEMENT

3. The agreement shall be in force for two (2) years from the date of signing of the agreement unless otherwise decided by ULB.
4. The agreement, at the discretion of ULB, is likely to be extended for another one (1) year subject to satisfactory performance by the SHG/TG Group/WPG/ Swachha Sathi(s).

II. ROLES AND RESPONSIBILITIES OF THE ULB

13. ULB shall designate a nodal officer to oversee the overall SWM.
14. ULB shall designate Sanitation Expert/Sanitation Intern who shall be responsible for monitoring the work of the Swachha Sathis.
15. ULB shall select the SHG/TG Group/WPGs for the deployment of Swachha Sathis.
16. ULB shall assign Swachha sathis to regularly monitor the door-to-door waste collection, to create awareness and training the citizens to ensure source segregation and ensure collection of segregated waste as per the SoP.
17. ULB shall conduct capacity building of Swachha Sathis on regular interval and provide handholding support to them as and when required.
18. ULB shall evaluate the performance of the SHG/TG Group/WPG and Swachha Sathis as per the performance indicators in the SoP.
19. ULB shall provide adequate resources for collection and transportation of Municipal Solid Waste.
20. ULB shall ensure timely payment to Swachha Sathis based on their performance.
21. ULB shall redress the grievances of the Swachha Sathis.

III. ROLES AND RESPONSIBILITIES OF THE SHG/TG GROUP/WPG

4. The SHG/TG Group/WPG shall submit the list of the eligible member(s) to be deployed as Swachha Sathis.
5. SHG/TG Group/WPG shall always ensure that the member(s) deployed as Swachha Sathis are trained by ULB before commencement of work.
6. The SHG/TG Group/WPG shall provide suitable replacement of Swachha Sathi (s) as prescribed in the SoP.

IV. ROLES AND RESPONSIBILITIES OF THE SWACHHA SATHI

4. Ensure door-to-door collection of segregated waste -1) wet waste, 2) dry waste, 3) E-waste & Domestic Hazardous waste.
5. Visit all the tagged Households/Commercial Units/Bulk Waste Generators including institutions and offices. and collect the User Fee.
6. Ensure the elimination of GVP
7. Facilitate meetings on sanitation with Federation/SDAs, Ward Level Committees at least once in a month.
8. Conduct IEC campaign on sanitation as mandated by ULB

9. While collecting the waste, demonstrate segregation of waste to the households not practicing segregation at source
10. Ensure that the route map is affixed on the Swachh Sabari and is followed by the Swachha Sabari Driver. Maintain a visit register for Households, Commercial units and Bulk Waste Generators including institutions and offices.
11. Maintain a register for complaints and share with Sanitation Intern/Nodal Officer.

V. GENERAL TERMS AND CONDITIONS

20. Binding Effect:

This Agreement shall be binding on and, to the extent such successors or assigns are permitted, inure to the benefit of the successors, and assigns of the ULB and the SHG/TG Group/WPG.

21. ULB's Right to Inspect:

The ULB or its designated / authorized representative / employees shall have the right to inspect the SWM status and records on user fee collection at any time or any portion thereof.

22. Amendment:

This Agreement may be amended only in writing duly agreed upon by the ULB and the SHG/TG Group/WPG.

23. Notices:

Any notice or other communication to be given by any Party to the other Party under or in connection with the matters contemplated by this Agreement shall be in writing and shall be delivered to persons as follows:

1. For the ULB:

Name:

Address:

Telephone:

E-mail:

2. For the SHG/TG Group/WPG:

Name:

Address:

Telephone:

E-mail:

24. Miscellaneous:

1. The terms of this Agreement shall not be construed to constitute a partnership, joint venture, or employer/employee relationship between the Parties.
2. This Agreement cannot be further sub-contracted; however, the SHG/TG Group/WPG can allow any member of the group to participate in training programme without any claim.
3. In the event of any provision of this Agreement being held or becoming invalid, unenforceable, or illegal for any reason, by both the Parties and a proceeding drawn to that effect and duly signed, this Agreement shall remain otherwise in full force apart from the said provision which shall be deemed to be deleted / modified, as the case may be. The Parties shall however attempt to replace the deleted provision with a legally valid provision that reflects the same purpose of the deleted provision to the greatest extent possible.
4. For non-adherence to any instructions / conditions / obligations arising out of the agreement, the ULB may issue written intimation to the SHG/TG Group/WPG clearly stating therein the violations made / aspects of non-observance, etc. and to rectify the same within a specified period to be indicated in the said written intimation. The SHG/TG Group/WPG should comply with the same and report to the ULB.

25. Applicable Law:

Applicable Law means the laws and any other instruments having the force of law in Odisha as they may be issued and in forced, from time to time.

26. Arbitration:

1. The Parties agree that the avoidance or early resolution of disputes is crucial for a smooth execution of the Agreement and the success of the assignment. The Parties shall use their best efforts to settle amicably all disputes arising out of or in connection with this Agreement or its interpretation.
2. Municipal Commissioner/ Project Director, District Urban Development Agency of the <district> shall act as the Mediator/Arbitrator for resolving any issue between the ULB and the SHG/TG Group/WPG within 30 days from date of receipt of the written complaint.
3. In case of failure in amicable settlement, the dispute shall be referred to the Collector of the district for redressal.
4. In case the dispute subsists, Parties may avail the forum under the Arbitration and Conciliation Act, 1996 read with Amendment Act, 2015 and shall be held in <Name of the town>, <Name of the district> India.

5. The Agreement and the rights and obligations of the Parties shall remain in full force and effect, pending the Award in the arbitration proceedings.

27. Force majeure:

Earthquakes, floods, fires, explosions, acts of God, acts of State, war, terrorism, action of any Government including changing governmental policy or any other situation, which is beyond the reasonable control of a Party may be treated as the Force Majeure event. For avoidance of doubt, it is clarified that, willful and intentional omissions, or commissions in relation to any Party shall not be considered as a Force Majeure event.

Neither of the Parties shall be entitled to claim benefit of a Force Majeure event unless it has intimated the other Party of occurrence of such event within 48 hours from the occurrence thereof, indicating there in the steps it is taking or intending to take to mitigate the effect of such Force Majeure event on performance of its obligations under this Agreement.

In the event of Force Majeure, the respective obligations and responsibilities shall remain suspended. If the Force Majeure continues for a period of more than 30 days, the Parties can re-negotiate this Agreement in good faith and if the Parties do not reach any consensus within the 15 days from the end of such 30 days period referred to above, this Agreement shall automatically stand terminated on such 45th day.

28. Recovery of cost for damage / loss of articles:

The ULB may recover the cost of items [equivalent to the market value of damaged or lost item(s)] or other assets if any in case of damage or loss.

29. Termination of Agreement:

The Agreement may be terminated by either party under the following situation:

1. In case the SHGs/TGs/WPGs is found to be violating the terms of the agreement or found not to be adhering to the terms during inspection by ULB, ULB shall have the right to terminate the Agreement by giving a written notice of not less than 30 days. During this period, the SHGs/TGs/WPGs shall be provided due opportunity to take corrective actions within 15 days from the date of notice with further extension, if deem/required not exceeding 7 days.
2. In case SHGs/TGs/WPGs expresses unwillingness/unable to continue with the Agreement, then the SHGs/TGs/WPGs shall have the right to get withdrawn from the Agreement on giving a 30 days' notice to ULB.
3. In case of dissolution of SHGs/TGs/WPGs, Agreement shall stand terminated on receipt of copy of the resolution stating the dissolution of

SHGs/TGs/WPGs.

4. The termination of the SHGs/TGs/WPGs shall be done by the ULB with due approval from the ULB level Swachhata Committee.

30. Replacement of Swachha Sathi:

1. Swachha Sathi may be replaced by the ULB by following due procedure on the ground of non-performance by them in two parameters i.e., User Fee Collection and GVP Free continuously for a period of 3 months.
2. Plant Manager may be replaced by the ULB by following due procedure on the ground of non-performance by them in all the parameters as detailed in the Incentive Structure mentioned in Table 3 continuously over a period of 3 months.
3. The names of the non-performing Swachha Sathis / Plant Managers will be notified to the concerned SHGs/TGs/WPGs in writing by the Nodal Officer of the Wealth Centre after observing their performance for a month. Based on the notice, handholding support is to be provided to the Swachha Sathis/ Plant Managers for improvement in their performance in the following month. If the non- performance continues even in the second month, the concerned nodal officer shall issue a written show cause notice to the SHGs/TGs/WPGs in the beginning of the third month and direct them to take needful action for improvement in the performance of the Swachha Sathi/ Plant Manager. Consistent non-compliance for the third month will lead to replacement of the Swachha Sathi/ Plant Manager. Failure on the part of the Nodal Officer of the Wealth Centre in notifying the SHGs/TGs/WPGs on the non- performance of Swachha Sathi/ Plant Manager shall be construed as serious lack of performance of the Nodal Officer and disciplinary action shall be taken by the Municipal Commissioner/ Executive Officer for the same.
4. If the non-performance of the concerned Swachha Sathis/ Plant Managers continues for a consecutive period of three months, the Nodal Officer of the Wealth Centre shall issue 7 days' written notice to SHGs/TGs/WPGs for replacement of the Swachha Sathi/ Plant Manager with due approval of the ULB Swachhata Committee and request the SHGs/TGs/WPGs to replace the concerned Swachha Sathis/ Plant Manager.
5. SHGs/TGs/WPGs shall provide a suitable replacement of the non-performing Swachha Sathi/ Plant Manager meeting the eligibility criteria within a period of seven working days with a written intimation to the ULB communicating the details of the Swachha Sathi, Plant Manager and the date of joining. In response, the ULBs shall communicate its acceptance or rejection seeking another candidate within seven days. Failure to provide suitable member within 15 days will make the Agreement with the

SHGs/TGs/WPGs liable for termination and ULB may take necessary steps for deployment of Swachha Sathi/ Plant Manager as the case may be from any other suitable SHGs/TGs/WPGs.

6. The ULB will impart capacity building training to the newly deployed Swachha Sathis/ Plant Managers for a minimum of two days before commencement of the work.
7. In case of any change in the number of Swachha Sathi/ Plant Manager, the Agreement shall be amended to reflect the updated number and details of Swachha Sathi/ Plant Manager deployed.

31. Indemnity:

To the fullest extent permitted by applicable law and professional regulations, the SHG/TG Group/WPG shall indemnify the ULB against all claims by third parties and resulting liabilities, losses, damages, costs, and expenses arising out of third party's use of the facilities.

32. Interpretation:

1. All words and expressions used in this Agreement shall, where relevant, have the meaning respectively assigned to it under any Act or any other law for the time being in force, and in the absence thereof, the Parties may suitably discuss and adhere to the modalities prescribed thereafter.
2. All words in singular shall be deemed to connote their respective plurals and vice-versa unless the context suggests otherwise.
3. The words "include" and "including" are to be construed without limitation.
4. The Appendixes to this Agreement form an integral part of this Agreement and shall be interpreted accordingly. However, the Appendix while providing information both relevant and with respect to certain aspects of the Agreement, shall be separate from it, and not read as integral to the Agreement.

Appendixes:

- 3) Incentive Framework (Appendix-1)
- 4) Monthly Report (Appendix-2)
- 5) Standard Operating Procedure, SOP (Appendix-3)

Appendix-1

Incentive Framework

Table-1: Monthly incentive framework for Swachha Sathi against primary performance indicator;

Sl. No.	Parameter	Eligibility	Incentive in Rs.	Means of Verification	Verification Responsibility	Replacement of Swachha Sathi
1	User Fee Collection on monthly demand from HHs/ Commercial	<50%	0	Ama Sahara App/ ULB Data	Sanitation Expert	Consistent non-performance in the 2 parameters over a period of 3 months.
		50-60%	2000	-do-	Sanitation Expert	
		61-80%	2000 + 1000	-do-	Sanitation Expert	
		> 80%	2000 + 2000	-do-	Sanitation Expert	
2	Garbage Vulnerable Points (GVP)	GVP Free	2000	Declaration by Swachha Sathi and validation of GVP status by Ward Level Committee (WLC) and submission to ULB by 27 th of each month. In case of non-receipt of report from WLC by the scheduled date, Officer in-charge of Sanitation shall verify the status for the concerned wards and release the incentive accordingly.	Officer in-charge of Sanitation	

Table-2: Additional monthly incentive framework for Swachha Sathi against secondary performance indicator;

Sl. No.	Parameter	Eligibility	Incentive in Rs.	Means of Verification	Beneficiary	Verification Responsibility
1	Quantity of Waste Collection (1 KG x No. of assigned HH/Com.x 25 Days)	> 80%	Rs.3000/- (@Rs.100 0/- each)	1. Daily Trip Register 2. Ama Sahara App	1. Swachha Sathi 2. Swachha Karmi-only Swachha Sabari Driver) 3. Swachha Karmi-(Swachha Sabari Helper only)	Sanitation Expert
2	Segregated Waste received in a month at the Wealth Centre	> 80%	Rs.3000/- (@Rs.100 0/- each)	Ama Sahara App or Daily Incoming Register	1. Swachha Sathi 2. Swachha Karmi (-Swachha Sabari Driver only) 3. Swachha Karmi (-Swachha Sabari Helper only)	Sanitation Expert
3	Door-step Sale of Mo Khata to Households, Commercial Establishments and Bulk Waste Generators including institutions and offices.		"Mo Khata" at 20% discounted rate	1. "Mo Khata" stock register 2. Ama Sahara App	1. Swachha Sathi 2. Swachha Karmi (-Swachha Sabari Driver only) 3. Swachha Karmi-(Swachha Sabari Helper only)	

AGREEMENT FOR DEPLOYMENT OF PLANT MANAGER(S)

This agreement, is made on this day, the _____ day of the month of _____, year, between <Name of ULB>, <District Name>, Odisha represented by the Commissioner or Executive Officer (As the case may be), here-in-after referred to as “1st Party”, which expression shall include its respective successors and permitted assignees of the part.

AND

<Name of the SHG/Transgender Group/Waste Picker Group>, <Ward No>, <ULB>, <District>, Odisha represented by the President/Secretary, here-in-after called the “2nd Party” which expression shall include its respective successors and permitted assignees.

WHEREAS, <ULB Name> is the Nodal Agency to implement Solid Waste Management in <ULB Name>. It is dedicated to improve collection and transportation of municipal waste in the city by involving Community Based Organizations (CBOs) to create livelihood opportunity for the urban SHG/TG Group/WPGs.

WHEREAS, the 1st Party intends to sign an agreement with 2nd Party to support collection and transportation of Municipal Solid Waste by deploying Plant Manager(s) for <Ward No__> as per the following terms and conditions;

I. DURATION OF THE AGREEMENT

5. The agreement shall be in force for two (2) years from the date of signing of the agreement unless otherwise decided by ULB.
6. The agreement, at the discretion of ULB, is likely to be extended for another one (1) year subject to satisfactory performance by the SHG/TG Group/WPG/ Plant Manager(s).

II. ROLES AND RESPONSIBILITIES OF THE ULB

22. ULB shall designate a nodal officer to oversee the overall SWM.

23. ULB shall designate Sanitation Expert/Sanitation Intern who shall be responsible for monitoring the work of the Plant Manager.
24. ULB shall select the SHG/TG Group/WPGs for the deployment of Plant Manager.
25. ULB shall assign Plant Manager to regularly monitor the door-to-door waste collection, to create awareness and training the citizens to ensure source segregation and ensure collection of segregated waste as per the SoP.
26. ULB shall conduct capacity building of Plant Manager on regular interval and provide handholding support to them as and when required.
27. ULB shall evaluate the performance of the SHG/TG Group/WPG and Plant Manager as per the performance indicators in the SoP.
28. ULB shall provide adequate resources for collection and transportation of Municipal Solid Waste.
29. ULB shall ensure timely payment to Plant Manager based on their performance.
30. ULB shall redress the grievances of the Plant Manager.

III. ROLES AND RESPONSIBILITIES OF THE SHG/TG GROUP/WPG

7. The SHG/TG Group/WPG shall submit the list of the eligible member(s) to be deployed as Plant Manager.
8. SHG/TG Group/WPG shall always ensure that the member(s) deployed as Plant Manager are trained by ULB before commencement of work.
9. The SHG/TG Group/WPG shall provide suitable replacement of Plant Manager(s) as prescribed in the SoP.

IV. ROLES AND RESPONSIBILITIES OF THE PLANT MANAGER

12. Ensure door-to-door collection of segregated waste -1) wet waste, 2) dry waste, 3) E-waste & Domestic Hazardous waste.
13. Visit all the tagged Households/Commercial Units/Bulk Waste Generators including institutions and offices. and collect the User Fee.
14. Ensure the elimination of GVP
15. Facilitate meetings on sanitation with Federation/SDAs, Ward Level Committees at least once in a month.
16. Conduct IEC campaign on sanitation as mandated by ULB
17. While collecting the waste, demonstrate segregation of waste to the households not practicing segregation at source
18. Ensure that the route map is affixed on the Swachh Sabari and is followed by the Swachha Sabari Driver. Maintain a visit register for Households, Commercial units and Bulk Waste Generators including institutions and offices.

19. Maintain a register for complaints and share with Sanitation Intern/Nodal Officer.

V. GENERAL TERMS AND CONDITIONS

33. Binding Effect:

This Agreement shall be binding on and, to the extent such successors or assigns are permitted, inure to the benefit of the successors, and assigns of the ULB and the SHG/TG Group/WPG.

34. ULB's Right to Inspect:

The ULB or its designated / authorized representative / employees shall have the right to inspect the SWM status and records on user fee collection at any time or any portion thereof.

35. Amendment:

This Agreement may be amended only in writing duly agreed upon by the ULB and the SHG/TG Group/WPG.

36. Notices:

Any notice or other communication to be given by any Party to the other Party under or in connection with the matters contemplated by this Agreement shall be in writing and shall be delivered to persons as follows:

1. For the ULB:

Name:

Address:

Telephone:

E-mail:

2. For the SHG/TG Group/WPG:

Name:

Address:

Telephone:

E-mail:

37. Miscellaneous:

1. The terms of this Agreement shall not be construed to constitute a partnership, joint venture, or employer/employee relationship between the Parties.

2. This Agreement cannot be further sub-contracted; however, the SHG/TG Group/WPG can allow any member of the group to participate in training programme without any claim.
3. In the event of any provision of this Agreement being held or becoming invalid, unenforceable, or illegal for any reason, by both the Parties and a proceeding drawn to that effect and duly signed, this Agreement shall remain otherwise in full force apart from the said provision which shall be deemed to be deleted / modified, as the case may be. The Parties shall however attempt to replace the deleted provision with a legally valid provision that reflects the same purpose of the deleted provision to the greatest extent possible.
4. For non-adherence to any instructions / conditions / obligations arising out of the agreement, the ULB may issue written intimation to the SHG/TG Group/WPG clearly stating therein the violations made / aspects of non-observance, etc. and to rectify the same within a specified period to be indicated in the said written intimation. The SHG/TG Group/WPG should comply with the same and report to the ULB.

38. Applicable Law:

Applicable Law means the laws and any other instruments having the force of law in Odisha as they may be issued and in force, from time to time.

39. Arbitration:

1. The Parties agree that the avoidance or early resolution of disputes is crucial for a smooth execution of the Agreement and the success of the assignment. The Parties shall use their best efforts to settle amicably all disputes arising out of or in connection with this Agreement or its interpretation.
2. Municipal Commissioner/ Project Director, District Urban Development Agency of the <district> shall act as the Mediator/Arbitrator for resolving any issue between the ULB and the SHG/TG Group/WPG within 30 days from date of receipt of the written complaint.
3. In case of failure in amicable settlement, the dispute shall be referred to the Collector of the district for redressal.
4. In case the dispute subsists, Parties may avail the forum under the Arbitration and Conciliation Act, 1996 read with Amendment Act, 2015 and shall be held in <Name of the town>, <Name of the district> India.
5. The Agreement and the rights and obligations of the Parties shall remain in full force and effect, pending the Award in the arbitration proceedings.

40. Force majeure:

Earthquakes, floods, fires, explosions, acts of God, acts of State, war, terrorism, action of any Government including changing governmental policy or any other situation, which is beyond the reasonable control of a Party may be treated as the Force Majeure event. For avoidance of doubt, it is clarified that, willful and intentional omissions, or commissions in relation to any Party shall not be considered as a Force Majeure event.

Neither of the Parties shall be entitled to claim benefit of a Force Majeure event unless it has intimated the other Party of occurrence of such event within 48 hours from the occurrence thereof, indicating there in the steps it is taking or intending to take to mitigate the effect of such Force Majeure event on performance of its obligations under this Agreement.

In the event of Force Majeure, the respective obligations and responsibilities shall remain suspended. If the Force Majeure continues for a period of more than 30 days, the Parties can re-negotiate this Agreement in good faith and if the Parties do not reach any consensus within the 15 days from the end of such 30 days period referred to above, this Agreement shall automatically stand terminated on such 45th day.

41. Recovery of cost for damage / loss of articles:

The ULB may recover the cost of items [equivalent to the market value of damaged or lost item(s)] or other assets if any in case of damage or loss.

42. Termination of Agreement:

The Agreement may be terminated by either party under the following situation:

1. In case the SHGs/TGs/WPGs is found to be violating the terms of the agreement or found not to be adhering to the terms during inspection by ULB, ULB shall have the right to terminate the Agreement by giving a written notice of not less than 30 days. During this period, the SHGs/TGs/WPGs shall be provided due opportunity to take corrective actions within 15 days from the date of notice with further extension, if deem/required not exceeding 7 days.
2. In case SHGs/TGs/WPGs expresses unwillingness/unable to continue with the Agreement, then the SHGs/TGs/WPGs shall have the right to get withdrawn from the Agreement on giving a 30 days' notice to ULB.
3. In case of dissolution of SHGs/TGs/WPGs, Agreement shall stand terminated on receipt of copy of the resolution stating the dissolution of SHGs/TGs/WPGs.
4. The termination of the SHGs/TGs/WPGs shall be done by the ULB with due

approval from the ULB level Swachhata Committee.

43. Replacement of Plant Manager:

1. Plant Manager may be replaced by the ULB by following due procedure on the ground of non-performance by them in two parameters i.e., User Fee Collection and GVP Free continuously for a period of 3 months.
2. Plant Manager may be replaced by the ULB by following due procedure on the ground of non-performance by them in all the parameters as detailed in the Incentive Structure mentioned in continuously over a period of 3 months.
3. The names of the non-performing Plant Managers will be notified to the concerned SHGs/TGs/WPGs in writing by the Nodal Officer of the Wealth Centre after observing their performance for a month. Based on the notice, handholding support is to be provided to the Plant Managers for improvement in their performance in the following month. If the non-performance continues even in the second month, the concerned nodal officer shall issue a written show cause notice to the SHGs/TGs/WPGs in the beginning of the third month and direct them to take needful action for improvement in the performance of the Plant Manager. Consistent non-compliance for the third month will lead to replacement of the Plant Manager. Failure on the part of the Nodal Officer of the Wealth Centre in notifying the SHGs/TGs/WPGs on the non-performance of Plant Manager shall be construed as serious lack of performance of the Nodal Officer and disciplinary action shall be taken by the Municipal Commissioner/ Executive Officer for the same.
4. If the non-performance of the concerned Plant Managers continues for a consecutive period of three months, the Nodal Officer of the Wealth Centre shall issue 7 days' written notice to SHGs/TGs/WPGs for replacement of the Plant Manager with due approval of the ULB Swachhata Committee and request the SHGs/TGs/WPGs to replace the concerned Plant Manager.
5. SHGs/TGs/WPGs shall provide a suitable replacement of the non-performing Plant Manager meeting the eligibility criteria within a period of seven working days with a written intimation to the ULB communicating the details of the Plant Manager and the date of joining. In response, the ULBs shall communicate its acceptance or rejection seeking another candidate within seven days. Failure to provide suitable member within 15 days will make the Agreement with the SHGs/TGs/WPGs liable for termination and ULB may take necessary steps for deployment of Plant Manager as the case may be from any other suitable SHGs/TGs/WPGs.
6. The ULB will impart capacity building training to the newly deployed Plant Managers for a minimum of two days before commencement of the work.
7. In case of any change in the number of Plant Manager, the Agreement shall be amended to reflect the updated number and details of Plant Manager deployed.

44. Indemnity:

To the fullest extent permitted by applicable law and professional regulations, the SHG/TG Group/WPG shall indemnify the ULB against all claims by third parties and resulting liabilities, losses, damages, costs, and expenses arising out of third party's use of the facilities.

45. Interpretation:

1. All words and expressions used in this Agreement shall, where relevant, have the meaning respectively assigned to it under any Act or any other law for the time being in force, and in the absence thereof, the Parties may suitably discuss and adhere to the modalities prescribed thereafter.
2. All words in singular shall be deemed to connote their respective plurals and vice-versa unless the context suggests otherwise.
3. The words "include" and "including" are to be construed without limitation.
4. The Appendixes to this Agreement form an integral part of this Agreement and shall be interpreted accordingly. However, the Appendix while providing information both relevant and with respect to certain aspects of the Agreement, shall be separate from it, and not read as integral to the Agreement.

Appendixes:

- 6) Incentive Framework (Appendix-1)
- 7) Monthly Report (Appendix-2)
- 8) Standard Operating Procedure, SOP (Appendix-3)

Annexure 21

Display Boards

Board -1: Daily Update on Waste for MCC



ସମ୍ପଦ କେନ୍ଦ୍ର, (Wealth Centre)
ଅଣୁ ଜୈବଜାତ ପ୍ରକ୍ରିୟାକରଣ କେନ୍ଦ୍ର, ପୋଖରୀପୁଟ
Micro Composting Centre (M.C.C), Pokhariput

ତାରିଖ Date	
ଆଜି ଏମ.ସି.ସିକୁ ଆସିଥିବା ଗାଡ଼ିଗୁଡ଼ିକର ମୋଟ ତ୍ରିପ ସଂଖ୍ୟା Total no of collection vehicles trip to M.C.C today:	
ମୋଟ ସଂଗୃହୀତ ଓଦା ଅଳିଆର ପରିମାଣ Total wet waste received today	କି.ଗ୍ରା kg
ଆଜି ଉତ୍ପାଦିତ ହୋଇଥିବା "ମୋ ଖତ" ର ପରିମାଣ Total compost generated today :	କି.ଗ୍ରା kg
ଆଜି ବିକ୍ରି ହୋଇଥିବା "ମୋ ଖତ" ର ପରିମାଣ Total compost sold today:	କି.ଗ୍ରା ଟଙ୍କା kg INR



Board -2: Daily Update on Waste for MRF



ସମ୍ପଦ କେନ୍ଦ୍ର (Wealth Centre) ସାମଗ୍ରୀ ପୁନର୍ଲାଭ କେନ୍ଦ୍ର, ପୋଖରୀପୁଟ Material Recovery Facility (M.R.F), Pokhariput		
ତାରିଖ Date		
ଏମ.ଆର.ଏଫ.କୁ ଆସିଥିବା ଗାଡ଼ି ଗୁଡ଼ିକର ମୋଟ ଟ୍ରିପ ସଂଖ୍ୟା Total no of collection vehicles trip to MRF today:		
ମୋଟ ସଂଗୃହିତ ଶୁଖିଲା ଅଳିଆର ପରିମାଣ Total dry waste received today	କି.ଗ୍ରା kg	
ମୋଟ ଅଲଗା କରାଯାଇଥିବା ଶୁଖିଲା ଅଳିଆର ପରିମାଣ Total dry waste sorted today:	କି.ଗ୍ରା kg	
ମୋଟ ପୁନଃବ୍ୟବହାରଯୋଗ୍ୟ ଅଳିଆ ବିକ୍ରି ପରିମାଣ Total recyclables sold today:	କି.ଗ୍ରା kg	ଟଙ୍କା INR

Board -3: Process flow of MRF





ବେଲିଙ୍ଗ ମେସିନରେ କାମ କଲା ବେଳେ ଜରୁରୀ ସାବଧାନତା ...

 ମେସିନରେ କାମ କରିବା ଏବଂ ଘରା କରିବା ସମୟରେ ଟିକର ବ୍ୟକ୍ତିଗତ ସୁରକ୍ଷା ଉପକରଣ (ପିପିଇ) ପିନ୍ଧନ୍ତୁ	 ଘରା ବା ମରାମତି କରିବା ପୂର୍ବରୁ ମେସିନର ପୁରା ବ୍ୟବହାର କରି ଦିଅନ୍ତୁ	 ୧ ମିଟର ଦୂରତା ଘରା ବା ମରାମତି କରିବା ପୂର୍ବରୁ ସମସ୍ତେ ମେସିନ ଠାରୁ ଦୂରରେ ରହନ୍ତୁ ଏବଂ ସୁରକ୍ଷିତ କରନ୍ତୁ	 ମେସିନରୁ ଛାତରେ ଘରା ନକରି ଘରା କରିବା ପାଇଁ ଉପଯୁକ୍ତ ଉପକରଣ ବ୍ୟବହାର କରନ୍ତୁ
 ଓପା ଛାତ ଡିମ୍ବା ଖାଲି ଛାତରେ ମେସିନ ଘରା କରନ୍ତୁ ନାହିଁ	 ଓପା ପାଦ ଡିମ୍ବା ଖାଲି ପାଦରେ ମେସିନକୁ ହୁଅନ୍ତୁ ନାହିଁ	 କାମ କଲାବେଳେ ଚାହାର ଯନ୍ତ୍ରପାତିରେ ଛାତ ଉଠାନ୍ତୁ ନାହିଁ	 ବେଶ ଖୋଲା ରଖି, ଡିଙ୍ଗା ପୋଷାକ, ଶାଢ଼ୀ, ଲମ୍ବା ଓକଣା ପିନ୍ଧି ମେସିନ ପାଖରେ କାମ କରନ୍ତୁ ନାହିଁ

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Board 5- Dos and Don'ts

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କନଭେୟର ବେଲ୍ଟରେ କାମ କଲାବେଳେ ଜୁରୁଟୀ ସାବଧାନତା ...

ମେସିନ୍ ଚାଲିଯାଇଲେ ଥିବା ମରଦା ସତ୍ତା ବନ୍ଦିବା ସମ୍ବନ୍ଧରେ ସୁରକ୍ଷା ଯୋଜନା କି ରାଜି ସୂଚିତ୍ କରନ୍ତୁ	ସତ୍ତା ବନ୍ଦିବା ବା ମରଦାଜି ବନ୍ଦିବା ପୂର୍ବରୁ ମେସିନ୍‌ର ସ୍ଥର ସାହାର ବନ୍ଦନ୍ତୁ
ବନ୍ଦରେଇର ବେଲ୍ଟରୁଟର ଉପରେ ବାହୁଆ ଲାଗିଥିଲେ କି ଏଥିରେ ସାମ ବଜନ୍ତୁ	ଜୁରୁଟୀ ବାହାରୁ ପ୍ରତିଷ୍ଠିତର ମେସିନ୍ ବନ୍ଦବନ୍ଦିବା ପାଇଁ ଦିଆଯାଉଥିବା ସୁରକ୍ଷା ସାଧନା ସତ୍ତା ବନ୍ଦ ବେଳେଇର ବାହାରବାହାରା ଅବସର ଥିବେ
ମେସିନ୍ ଆଗର ବୋଉରେ ଦିଆ ଲାଗ ବୋଉରେ ବୁଜି ସାମ ବେଳ୍ଟର ଉପରୁ ବୁଜନ୍ତୁ	ବେଳ୍ଟର ଉପରୁ ବୁଜି ବୋଉରେ ବୁଜି ସାମ ବେଳ୍ଟର ଉପରୁ ବୁଜନ୍ତୁ

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ଓଡ଼ିଶା ସରକାର



ଜାତୀୟ ନଗରୀୟ ସ୍ୱଚ୍ଛତା କମିଶନ



ଜାତୀୟ ନଗରୀୟ ସ୍ୱଚ୍ଛତା କମିଶନ



ଜାତୀୟ ନଗରୀୟ ସ୍ୱଚ୍ଛତା କମିଶନ

ଶୁଦ୍ଧତା ମେସିନରେ କାମ କଲା ବେଳେ ଜୁରୁରୀ ସାଧାଧାନତା ...

✔


ଶେସିନରେ କାମ କରିବା ଏବଂ ସଜା କରିବା ସମୟରେ ମୁହଁରୁ ଲୁଗା ଢାଳିବା (ପିଟିବା) ପିନ୍ଧିବା

✔


ସଜା ବା ଗଲ୍ୟାନ୍ କରିବା ସ୍ୱଚ୍ଛତା ମେସିନର ସ୍ୱଚ୍ଛତା କରାଯିବା ପରେ କରାଯିବ

✔


ଏ ମିଟର ଦୂରରେ ରହନ୍ତୁ

✔


ଶେସିନରୁ ଛାଡ଼ିବା ପରେ ଶେସିନରୁ ସଜା କରିବା ପାଇଁ ଉପଯୁକ୍ତ ଉପକରଣ ବ୍ୟବହାର କରନ୍ତୁ

✗


ଜଳ ବାହାରିବା ଥିବା ସମୟରେ ଶେସିନରୁ ସଜା କରନ୍ତୁ ନାହିଁ

✗


ଜଳ ବାହାରିବା ଥିବା ସମୟରେ ଶେସିନରୁ ଛାଡ଼ିବା କରନ୍ତୁ ନାହିଁ

✗


କାମ କରୁଥିବା ସମୟରେ ଶେସିନରୁ ଛାଡ଼ିବା କରନ୍ତୁ ନାହିଁ

✗


କେବଳ ଶେସିନରୁ ଛାଡ଼ିବା ପରେ, ଶେସିନରୁ ଛାଡ଼ିବା ପରେ ଶେସିନରୁ ଛାଡ଼ିବା କରନ୍ତୁ ନାହିଁ

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“Sanitation is more important than independence”

Mahatma Gandhi

**Housing and Urban Development Department
Government of Odisha**